## Supplementary Materials

**Table S1.** Country classification used for group-wise analysis.

| Sr. No. | Country | Economy group | Classification basis | Analytical use |
|---|---|---|---|---|
| **1** | Australia | High-income economy | Economy-group classification used in this study | Group-wise descriptive comparison and scenario sensitivity |
| **2** | Austria | | | |
| **3** | Belgium | | | |
| **4** | Canada | | | |
| **5** | Czech Republic | | | |
| **6** | Denmark | | | |
| **7** | Estonia | | | |
| **8** | Finland | | | |
| **9** | France | | | |
| **10** | Germany | | | |
| **11** | Greece | | | |
| **12** | Hungary | | | |
| **13** | Iceland | | | |
| **14** | Ireland | | | |
| **15** | Israel | | | |
| **16** | Italy | | | |
| **17** | Japan | | | |
| **18** | Korea | | | |
| **19** | Latvia | | | |
| **20** | Lithuania | | | |
| **21** | Luxembourg | | | |
| **22** | Netherlands | | | |
| **23** | New Zealand | | | |
| **24** | Norway | | | |
| **25** | Portugal | | | |
| **26** | Slovak Republic | | | |
| **27** | Slovenia | | | |
| **28** | Spain | | | |
| **29** | Sweden | | | |
| **30** | Switzerland | | | |

| | | | | |
|---|---|---|---|---|
| **31** | United Kingdom | | | |
| **32** | United States | | | |
| **33** | Chile | Emerging economy | | |
| **34** | Colombia | | | |
| **35** | Costa Rica | | | |
| **36** | Mexico | | | |
| **37** | Türkiye | | | |

**Note:** The analytical sample comprises 32 economies classified as high-income and 5 classified as emerging economies for the purposes of group-wise descriptive comparison and scenario sensitivity analysis. Economy-group classification is used as an analytical grouping variable and should not be interpreted as a direct measure of LTC-system maturity, institutional quality, financing sustainability, or policy performance. Because the emerging-economy subgroup contains only five economies, group-specific results are interpreted as exploratory.

**Table S2.** Missing-data summary and analytical treatment for the harmonized country-year panel.

| Sr. No. | Variable | Analytical role | Expected country-year observations | Available observations | Missing observations | Missing (%) | Countries with available data | Observed year range | Recommended treatment |
|---|---|---|---|---|---|---|---|---|---|
| **1** | HealthExp_GDP / health expenditure burden | Primary expenditure outcome | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Retain directly; complete coverage and no missing-value treatment required |
| **2** | Gov_Share | Supplementary public-financing indicator | 518 | 41 | 477 | 92.08 | 3 | 2010–2023 | Do not impute to the full panel; use only in exploratory restricted-sample financing analyses |
| **3** | Pop65 | Population-aging indicator | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Retain directly as a core predictor; no interpolation or imputation required |
| **4** | Dependency | Demographic-pressure indicator | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Retain directly as a core predictor; no interpolation or imputation required |
| **5** | LTC_Workers | Supplementary LTC workforce-capacity indicator | 518 | 58 | 460 | 88.80 | 16 | 2020–2023 | Do not impute to the full panel; use only in exploratory restricted-sample workforce analyses |
| **6** | GDP_pc | Macroeconomic-capacity indicator | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Retain directly as a core predictor; no interpolation or imputation required |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **7** | Public_Exposure | Derived public-financing exposure indicator | 518 | 41 | 477 | 92.08 | 3 | 2010–2023 | Construct only where both HealthExp_GDP and Gov_Share are observed; treat as a supplementary restricted-sample descriptive indicator |
| **8** | FPI | Fiscal Pressure Index | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Construct from HealthExp_GDP and Dependency; use as a derived descriptive or sensitivity indicator rather than an independent measure of fiscal sustainability |
| **9** | Worker_Ratio | Workforce adequacy ratio | 518 | 58 | 460 | 88.80 | 16 | 2020–2023 | Construct only where LTC_Workers and Pop65 are observed; use as a supplementary restricted-sample workforce indicator |
| **10** | Affordability | Macroeconomic affordability ratio | 518 | 518 | 0 | 0.00 | 37 | 2010–2023 | Construct from GDP_pc and HealthExp_GDP; use as a derived descriptive or sensitivity indicator |

**Note:** The expected panel size comprises 37 economies observed annually from 2010 to 2023, yielding 518 country-year observations. In the final harmonized analytical panel, HealthExp_GDP, Pop65, Dependency, and GDP_pc have complete coverage; therefore, no interpolation or grouped imputation is required for the core econometric and machine-learning variables. Government financing share and LTC workforce capacity remain substantially incomplete and are not imputed to create full-panel observations. Analyses involving Gov_Share, LTC_Workers, Public_Exposure, or Worker_Ratio are restricted to country-years with observed parent variables and are interpreted as exploratory supplementary evidence. FPI and Affordability are deterministic transformations of completely observed core variables and are used for descriptive interpretation or sensitivity analysis rather than as independent measures of fiscal or LTC financing sustainability. Exact variable-specific missing-data treatment is further summarized in Supplementary Table S2C.

**Table S2A.** Source-variable clarification and interpretation of study variables

| Manuscript variable | Clarified variable name | Source-variable meaning | Unit | Analytical role | Interpretation |
|---|---|---|---|---|---|
| **HealthExp_GDP** | Health expenditure burden | Current health expenditure as a share of GDP; World Bank WDI indicator SH.XPD.CHEX.GD.ZS, sourced from the WHO | % of GDP | Primary outcome | Measures total health-system expenditure relative to national economic output. It is interpreted as health expenditure burden and not as direct LTC expenditure, LTC financing sustainability, or fiscal solvency. |

| | | | | | |
|---|---|---|---|---|---|
| | | Global Health Expenditure Database. | | | |
| **Gov_Share** | Government financing share | Public/government share of health financing | % or proportion | Supplementary restricted-sample financing indicator | Describes the public-sector financing share where observed. Because coverage is limited to 41 country-year observations from 3 countries, it is not used as a core full-panel predictor and does not support general cross-country financing conclusions. The archived series is retained only as restricted-sample supplementary evidence; missing observations are not imputed to create a full-panel predictor. |
| **Pop65** | Population aged 65 years and above | Population ages 65 and above as a percentage of total population; World Bank WDI indicator SP.POP.65UP.TO.ZS (UN Population Division source). | % of total population | Core demographic predictor | Represents population aging and is used to examine its association with and predictive contribution to health expenditure burden. It should not be interpreted as a direct measure of LTC need. |
| **Dependency** | Total age dependency ratio | Dependents aged 0–14 and 65+ relative to the working-age population; World Bank WDI indicator SP.POP.DPND. | % of working-age population | Core demographic predictor | Represents demographic dependency pressure. It is used in the core econometric and predictive models and as a component of the derived Fiscal Pressure Index. |
| **GDP_pc** | Gross domestic | GDP per capita; | Current US$; | Core macroeconomic predictor | Represents macroeconomic capacity and is used in expenditure-burden and income- |

| | | | | | |
|---|---|---|---|---|---|
| | product per capita | benchmark World Bank WDI indicator NY.GDP.PCAP.CD (current US$), with robustness using NY.GDP.PCAP.PP.KD (PPP, constant 2021 international $). | robustness in constant 2021 international $ | | elasticity analyses. It is also a component of the derived affordability ratio. |
| **LTC_Workers** | Long-term care workforce capacity | Formal LTC workforce availability reported in the source data | Count, rate, or source-specific workforce measure | Supplementary restricted-sample workforce indicator | Provides exploratory evidence on formal care-service capacity. Coverage is limited to 58 country-year observations from 16 countries, so it is not used as a core full-panel predictor. The source-specific workforce definition and limited country coverage preclude full-panel generalization. |
| **Public_Exposure** | Public financing exposure | Health expenditure burden combined with government/public financing share | Derived percentage/proportion | Supplementary derived financing indicator | Constructed only where Gov_Share is observed. It provides an exploratory description of public financing exposure and is not interpreted as direct public LTC expenditure or an independent measure of fiscal sustainability. |
| **FPI** | Fiscal Pressure Index | Health expenditure burden combined with demographic dependency | Derived index | Derived descriptive/sensitivity indicator | Deterministic combination of health expenditure burden and dependency pressure. It is used for descriptive or sensitivity analysis and not as an independent measure of fiscal sustainability. |
| **Worker_Ratio** | Workforce adequacy ratio | LTC workforce relative to the older population | Derived ratio | Supplementary derived workforce indicator | Provides an exploratory measure of formal workforce availability relative to the older population. Its coverage is inherited from LTC_Workers and therefore remains restricted. |

| | | | | | |
|---|---|---|---|---|---|
| **Affordability** | Affordability ratio | GDP per capita relative to health expenditure burden | Derived ratio | Derived descriptive/sensitivity indicator | Provides a descriptive measure of macroeconomic capacity relative to health expenditure burden. Because it is deterministically constructed from existing variables, it is not an independent measure of affordability, solvency, or fiscal sustainability. |

**Note:** HealthExp_GDP, Pop65, Dependency, and GDP_pc constitute the complete-coverage core variables used in the principal econometric and predictive analyses. Gov_Share and LTC_Workers have substantially restricted country-year coverage and are therefore used only in supplementary exploratory analyses. Public_Exposure, FPI, Worker_Ratio, and Affordability are deterministic transformations of observed variables and are treated as descriptive or sensitivity indicators rather than independent measures of long-term care financing or fiscal sustainability.

**Table S2B.** Coverage profile and interpretation limits of LTC-specific financing and workforce indicators.

| Variable | Available country-year observations | Unique countries with available data | Full panel reference | Coverage issue | Analytical treatment | Interpretation limit |
|---|---|---|---|---|---|---|
| **Gov_Share** | 41 | 3 | 518 country-years from 37 countries | Very limited country coverage | Supplementary financing indicator; used cautiously in derived public-exposure analysis | Does not support broad group-level financing claims across all countries |
| **LTC_Workers** | 58 | 16 | 518 country-years from 37 countries | Limited and uneven country-year coverage | Supplementary workforce-capacity indicator; used in restricted and sensitivity analysis | Does not support definitive cross-country workforce conclusions |
| **Public_Exposure / public financing exposure** | Depends on Gov_Share availability | Depends on Gov_Share availability | 518 country-years from 37 countries | Inherits missingness from Gov_Share | Supplementary derived indicator | Interpreted as exploratory public financing exposure, not a full-panel estimate |
| **Worker_Ratio** | Depends on LTC_Workers and Pop65 availability | Depends on LTC_Workers' availability | 518 country-years from 37 countries | Inherits missingness from LTC_Workers | Supplementary derived workforce indicator | Interpreted as exploratory service-capacity evidence |

**Table S2C.** Variable-specific missing-data treatment in the final analytical panel.

| Variable | Final available observations | Interpolated observations | Group-imputed observations | Missing retained | Analytical treatment |
|---|---|---|---|---|---|
| **HealthExp_GDP** | 518 | 0 | 0 | 0 | Primary outcome; complete panel |
| **Pop65** | 518 | 0 | 0 | 0 | Core predictor; complete panel |
| **Dependency** | 518 | 0 | 0 | 0 | Core predictor; complete panel |

| | | | | | |
|---|---|---|---|---|---|
| **GDP_pc** | 518 | 0 | 0 | 0 | Core predictor; complete panel |
| **Gov_Share** | 41 | 0 | 0 | 477 | Restricted-sample exploratory analysis only |
| **LTC_Workers** | 58 | 0 | 0 | 460 | Restricted-sample exploratory analysis only |

Note: HealthExp_GDP, Pop65, Dependency, and GDP_pc are complete across all 518 country-year observations; interpolated observations = 0 and group-imputed observations = 0 for every core variable. Missing government-financing and LTC-workforce observations were not filled to construct full-panel predictors. Analyses involving those variables use only country-years for which the corresponding observations are available.

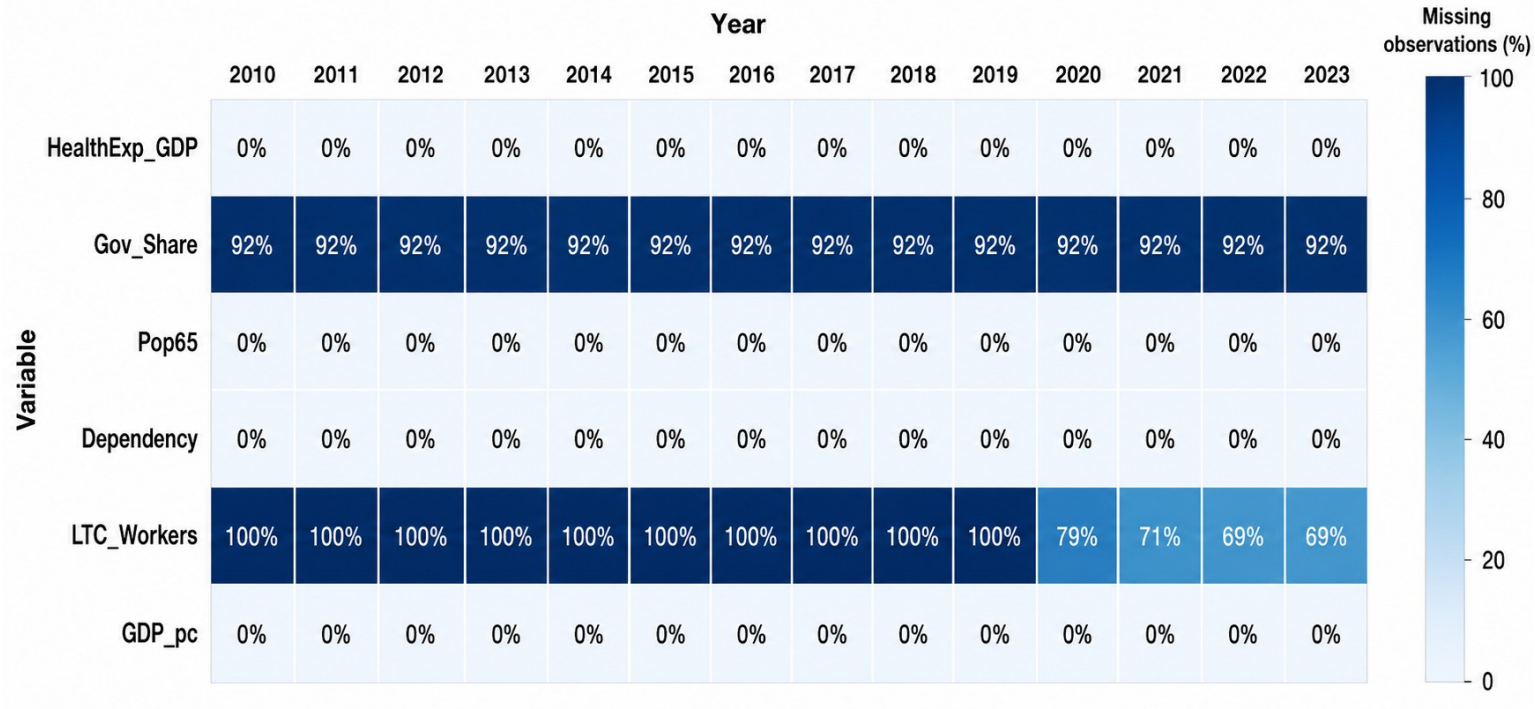


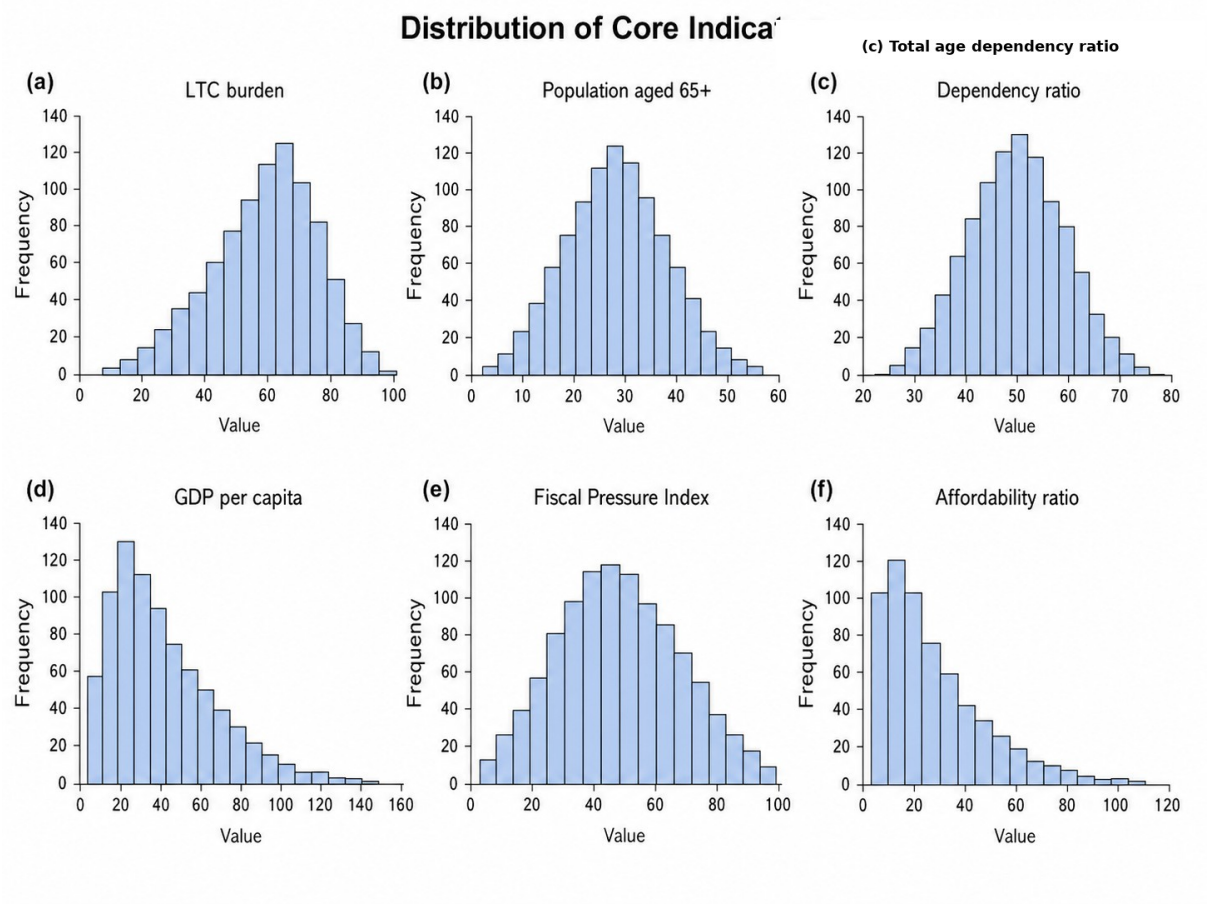

**Figure S1.** Missing-data pattern across variables and years.

Figure S1 shows the annual missing-data pattern for the core and supplementary variables across 2010–2023. Health expenditure burden, population aged 65 years and above, total age dependency ratio, and GDP per capita have complete coverage in the final 37-economy panel. Government-financing and LTC-workforce indicators remain restricted-coverage supplementary variables and are not imputed to create full-panel predictors.

**Table S3. Robustness strategy for core and restricted-sample analyses.**

| Robustness model | Sample specification | Missing-data treatment | Variables included | Purpose | Interpretation |
|---|---|---|---|---|---|
| **R1** | Core full analytical panel | No imputation required | HealthExp_GDP, Pop65, Dependency, GDP_pc, year/economy-group information as applicable | Benchmark econometric and predictive specification | Uses the complete-coverage 518-observation panel |
| **R2** | Core-variable specification | No imputation required | HealthExp_GDP, Pop65, Dependency, GDP_pc | Confirms results using only variables with complete coverage | Provides the principal full-panel demographic and macroeconomic evidence |
| **R3** | Restricted-sample financing description | Missing Gov_Share observations retained as missing; no full-panel imputation | Core variables + Gov_Share and related observed financing measures | Examines whether observed financing information provides additional exploratory evidence | Restricted to 41 observations from 3 countries; not generalized to the full panel |
| **R4** | Restricted-sample workforce description | Missing LTC_Workers observations retained as missing; no full-panel imputation | Core variables + LTC_Workers and Worker_Ratio where observed | Examines workforce-capacity information | Restricted to 58 observations from 16 countries; exploratory only |
| **R5** | High-income subsample | Core complete-coverage variables retained directly | Core variables | Examines whether full-panel relationships differ within the high-income group | Subgroup evidence; not a causal comparison |
| **R6** | Emerging-economy subsample | Core complete-coverage variables retained directly | Core variables | Examines whether relationships differ within the emerging-economy group | Exploratory because only five emerging economies are represented |
| **R7** | Alternative derived-outcome analysis | Parent-variable observations used directly | Public_Exposure, FPI, Affordability, or Worker_Ratio as applicable | Evaluates descriptive sensitivity to alternative constructed measures | Derived measures are not independent indicators of fiscal sustainability |
| **R8** | Scenario sensitivity checks | No imputation of financing or workforce variables | Pop65, Dependency, GDP_pc | Tests sensitivity to demographic/GDP trend multipliers and projection support | 2024–2028 sensitivity analysis only; not a policy-effect or long-term forecast |

The core specification therefore requires no imputation-based sensitivity comparison. Government-financing and LTC-workforce variables remain missing where unavailable and are used only in restricted-sample exploratory analyses.




Supplementary Figure S2. Histograms of core demographic, fiscal, and macroeconomic indicators.

Figure S2 presents the distribution of key variables across a broad range of countries and years, including the health expenditure burden, the elderly population share, the total age dependency ratio, gross domestic product per capita, the Fiscal Pressure Index, and the affordability ratio. These histograms support the assessment of skewness, dispersion, and potential outliers prior to econometric and machine-learning modeling.

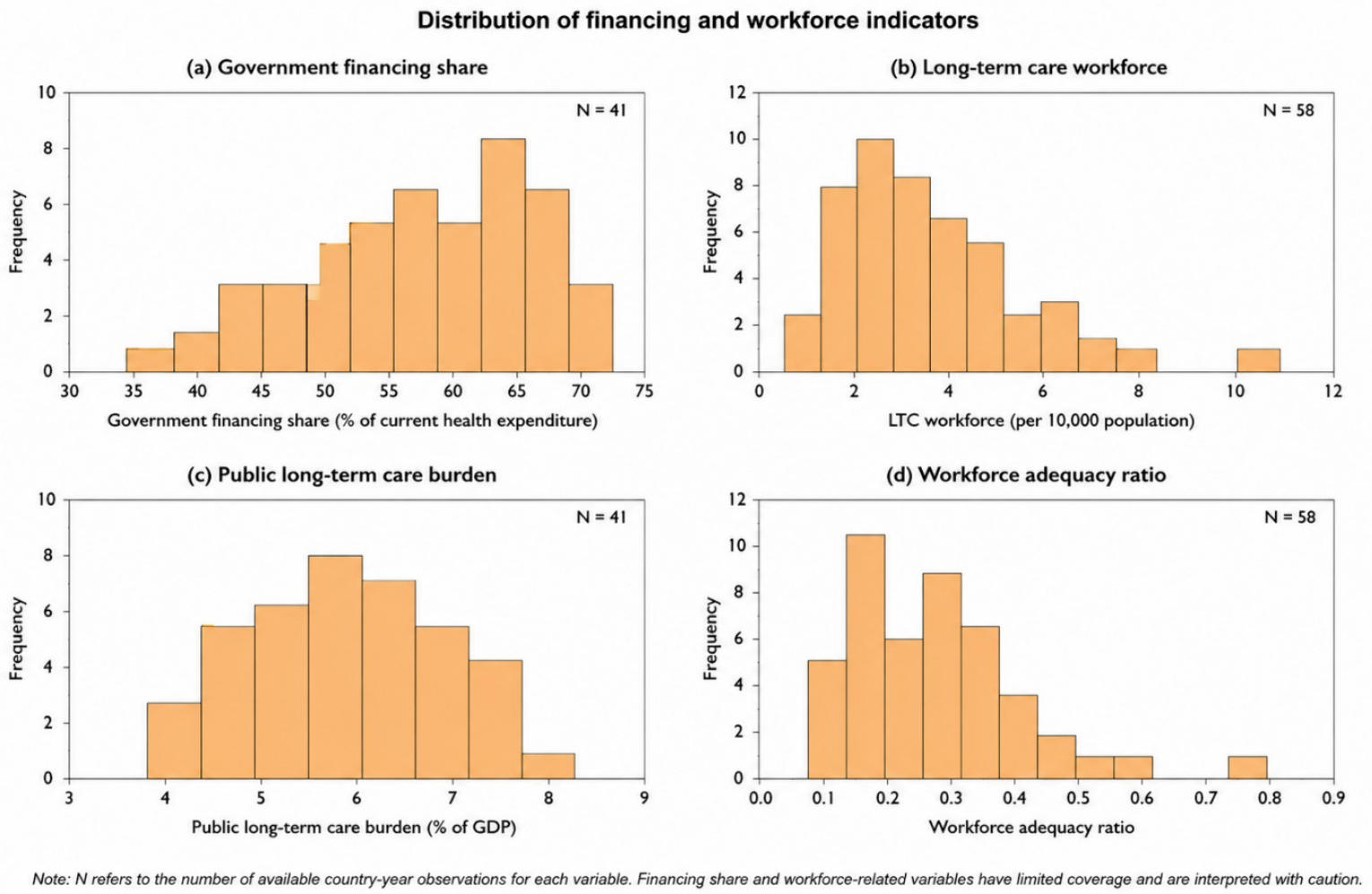



Figure S3. Histograms of financing and workforce-related indicators.

Figure S3 depicts the distribution of government financing share, long-term care workforce capacity, public financing exposure, and workforce adequacy ratio. These variables are shown separately because they have more limited country-year coverage and require cautious interpretation in the main analysis.

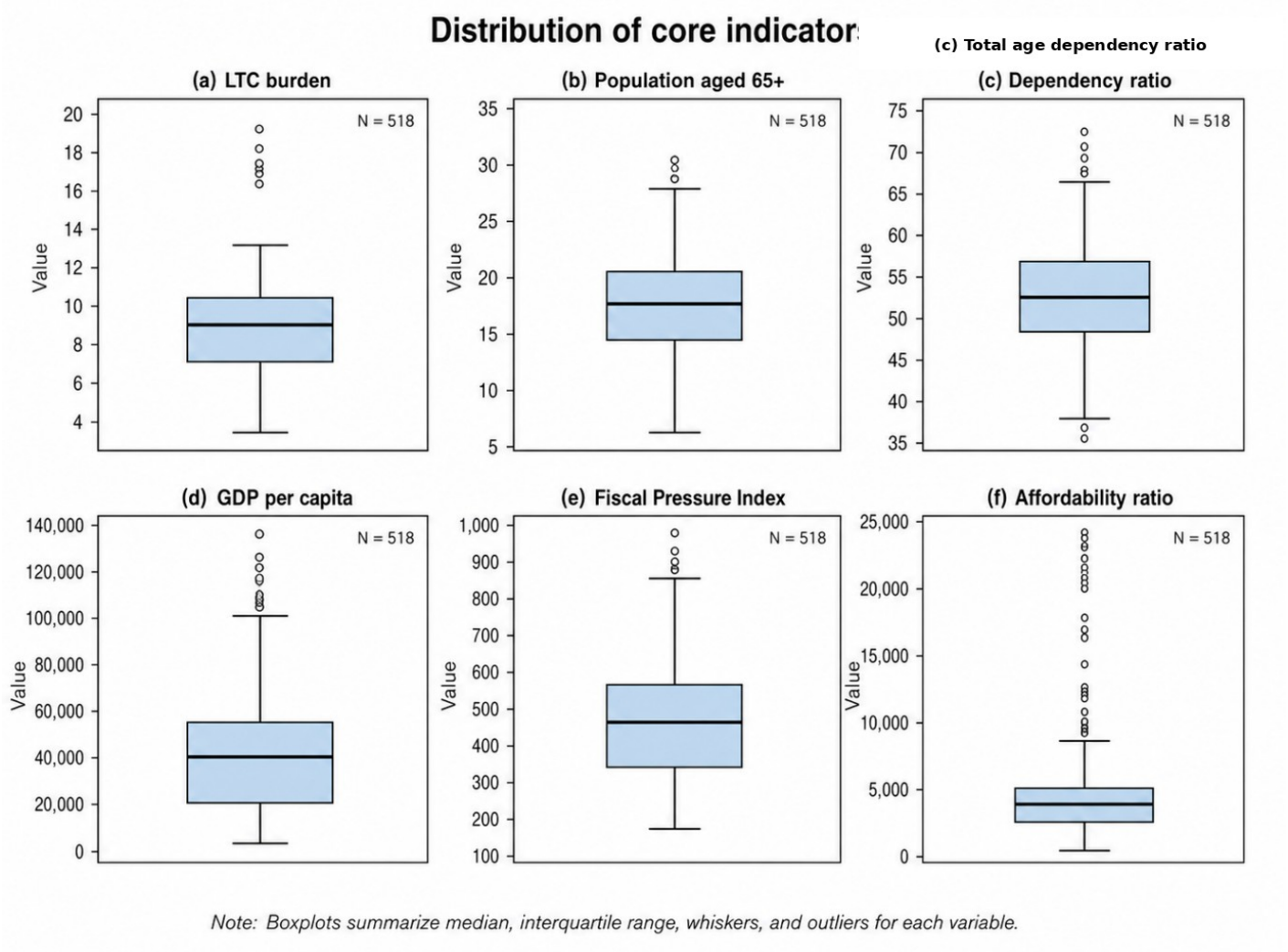



Figure S4. Boxplots of core demographic, fiscal, and macroeconomic indicators.

Figure S4 documents the distributional spread, cross-country dispersion, and potential outliers of the core indicators. Predictive model performance is evaluated separately using expanding-window validation.

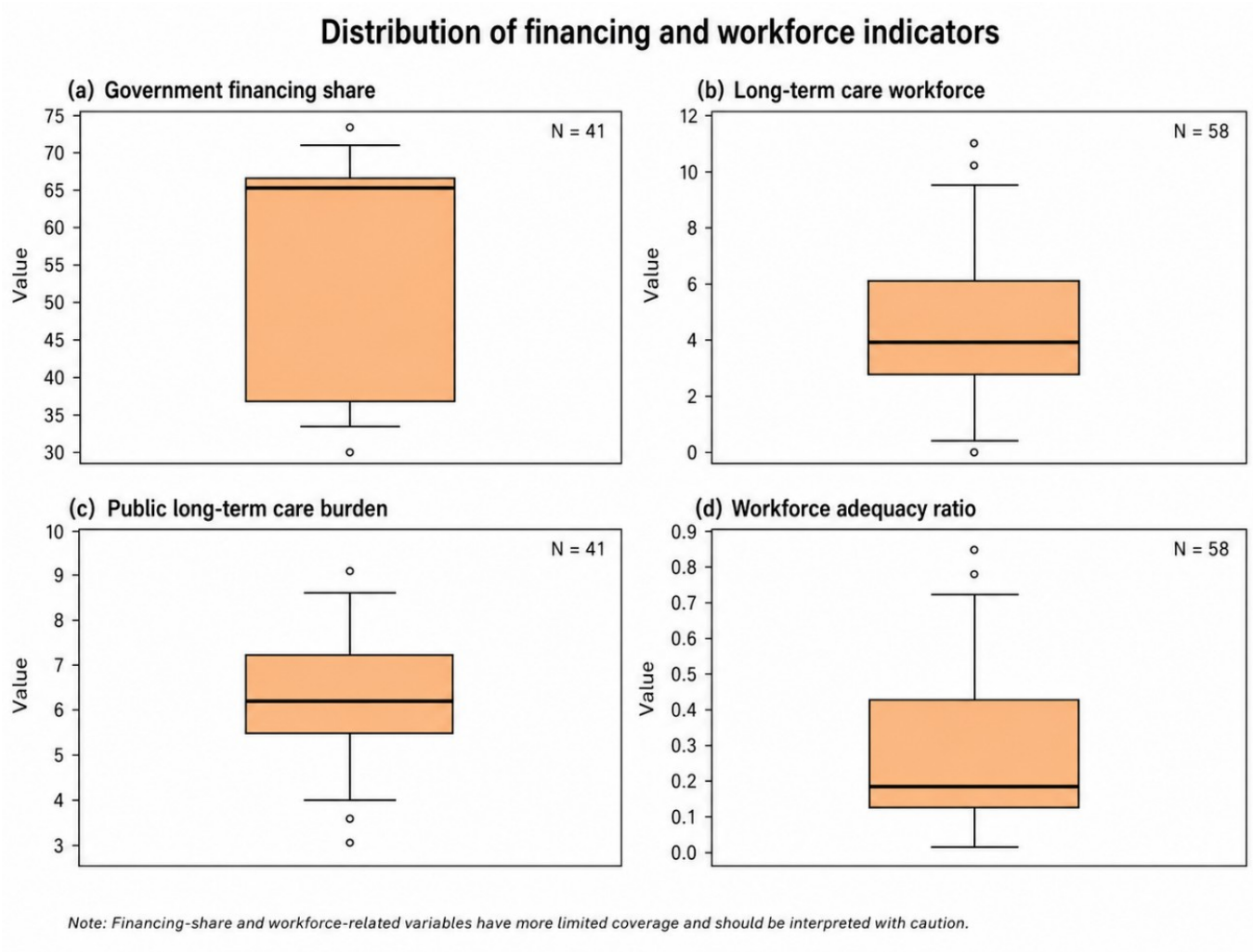



Figure S5. Boxplots of financing and workforce-related indicators.

Figure S5 presents the boxplots for government financing share, long-term care workforce capacity, public financing exposure, and workforce adequacy ratio. These indicators show clear dispersion and outlier patterns, while their low observation counts support the cautious interpretation and the robustness checks applied in the modeling framework.

**Table S4.** Full correlation matrix of key study variables

Note: This matrix is retained only as supplementary descriptive material. Correlations among the four core variables use the complete 518-observation panel. Correlations involving government financing and LTC-workforce measures rely on much smaller available samples (41 and 58 observations, respectively), while correlations involving deterministic derived indicators are partly mechanical. These coefficients should not be compared as if they were estimated from a common sample.

| Variable | Health expenditure burden | Gov. share | Pop. 65+ | Dependency | GDP per capita | LTC workforce | Public financing exposure | FPI | Worker ratio | Affordability |
|---|---|---|---|---|---|---|---|---|---|---|
| **Health expenditure burden** | 1.00 | -0.87 | 0.42 | 0.31 | 0.33 | -0.11 | 0.34 | 0.94 | -0.23 | -0.09 |
| **Gov. share** | -0.87 | 1.00 | -0.24 | -0.59 | -0.62 | 0.11 | 0.15 | -0.87 | 0.11 | -0.18 |
| **Pop. 65+** | 0.42 | -0.24 | 1.00 | 0.56 | 0.16 | -0.24 | 0.78 | 0.56 | -0.50 | -0.00 |
| **Dependency** | 0.31 | -0.59 | 0.56 | 1.00 | 0.10 | 0.28 | 0.44 | 0.61 | 0.23 | -0.06 |
| **GDP per capita** | 0.33 | -0.62 | 0.16 | 0.10 | 1.00 | 0.40 | 0.58 | 0.31 | 0.42 | 0.88 |
| **LTC workforce** | -0.11 | 0.11 | -0.24 | 0.28 | 0.40 | 1.00 | 0.93 | 0.00 | 0.93 | 0.32 |
| **Public financing exposure** | 0.34 | 0.15 | 0.78 | 0.44 | 0.58 | 0.93 | 1.00 | 0.34 | 0.93 | 0.75 |
| **FPI** | 0.94 | -0.87 | 0.56 | 0.61 | 0.31 | 0.00 | 0.34 | 1.00 | -0.13 | -0.08 |
| **Worker ratio** | -0.23 | 0.11 | -0.50 | 0.23 | 0.42 | 0.93 | 0.93 | -0.13 | 1.00 | 0.40 |
| **Affordability** | -0.09 | -0.18 | -0.00 | -0.06 | 0.88 | 0.32 | 0.75 | -0.08 | 0.40 | 1.00 |

Note: The Fiscal Pressure Index is mechanically related to the health expenditure burden because it is constructed from the health expenditure burden and the total age dependency ratio. Therefore, its high correlation with the primary outcome is expected and should not be interpreted as an independent empirical association. FPI is used as a derived fiscal-pressure indicator and as an alternative robustness outcome, not as an independent predictor of the health expenditure burden in the core model.

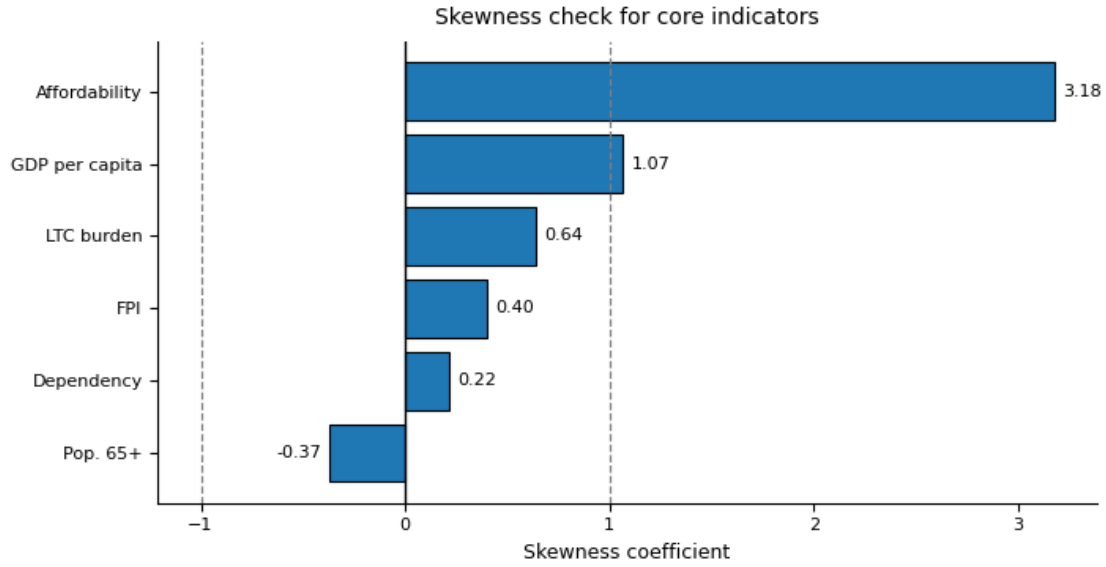



**Figure S6.** Skewness check for core demographic, fiscal, and macroeconomic indicators.

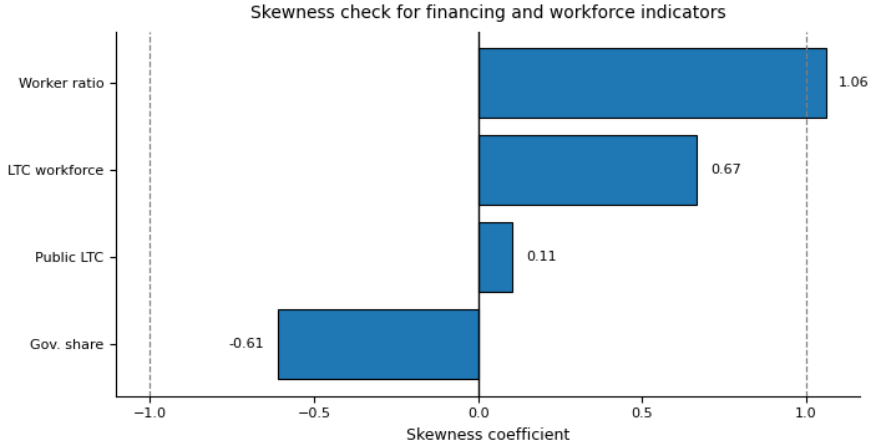



**Figure S7.** Skewness check for financing and workforce-related indicators.

**Table S5.** Full pooled log-log model results with country-clustered standard errors

| Variable / model component | Coefficient | Clustered SE | t statistic | p-value | 95% CI |
|---|---|---|---|---|---|
| **Population aged 65+, elasticity** | 0.254 | 0.127 | 2.006 | 0.052 | −0.003 to 0.511 |
| **Total age total age dependency ratio, elasticity** | 0.287 | 0.306 | 0.939 | 0.354 | −0.333 to 0.908 |
| **GDP per capita, elasticity of expenditure burden** | 0.134 | 0.081 | 1.659 | 0.106 | −0.030 to 0.297 |
| **Year effects** | Included | — | — | — | — |
| **Country effects** | Not included | — | — | — | — |
| **Country-clustered inference** | Yes | — | — | — | — |
| **Observations** | 518 | — | — | — | — |
| **Countries** | 37 | — | — | — | — |

Note: The dependent variable is the natural logarithm of health expenditure burden, measured as total health expenditure as a percentage of GDP. Continuous predictors are entered in natural logarithms; therefore, the reported coefficients are elasticities of the expenditure-to-GDP burden. Standard errors are clustered by country. Statistical inference uses 37 country clusters, with p-values and confidence intervals based on the corresponding cluster-level degrees of freedom. The aging coefficient is positive but is not statistically significant at the 5% level.

**Table S6.** Full fixed-effects log-log model results with country-clustered standard errors

| Variable / model component | Coefficient | Clustered SE | t statistic | p-value | 95% CI |
|---|---|---|---|---|---|
| **Population aged 65+, elasticity** | 0.030 | 0.323 | 0.094 | 0.926 | −0.624 to 0.685 |
| **Total age total age dependency ratio, elasticity** | 0.197 | 0.214 | 0.917 | 0.365 | −0.238 to 0.632 |
| **GDP per capita, elasticity of expenditure burden** | −0.143 | 0.145 | −0.988 | 0.330 | −0.438 to 0.151 |
| **Country fixed effects** | Included | — | — | — | — |
| **Year effects** | Included | — | — | — | — |
| **Country-clustered inference** | Yes | — | — | — | — |
| **Observations** | 518 | — | — | — | — |
| **Countries** | 37 | — | — | — | — |

Note: The dependent variable and continuous predictors are expressed in natural logarithms. The fixed-effects specification controls for time-invariant country characteristics and common year-specific effects. Standard errors are clustered by country. The coefficients therefore characterize conditional within-country relationships after accounting for country and year effects. None of the three core elasticities is statistically distinguishable from zero in this specification.

**Table S7.** Full random-effects log-log model results with country-clustered standard errors

| Variable / model component | Coefficient | Clustered SE | t statistic | p-value | 95% CI |
|---|---|---|---|---|---|
| **Population aged 65+, elasticity** | 0.164 | 0.213 | 0.772 | 0.445 | −0.267 to 0.595 |
| **Total age total age dependency ratio, elasticity** | 0.208 | 0.207 | 1.005 | 0.322 | −0.211 to 0.627 |
| **GDP per capita, elasticity of expenditure burden** | −0.088 | 0.122 | −0.719 | 0.477 | −0.335 to 0.160 |

| | | | | | |
|---|---|---|---|---|---|
| **Country random intercept** | Included | — | — | — | — |
| **Year effects** | Included | — | — | — | — |
| **Country-clustered inference** | Yes | — | — | — | — |
| **Observations** | 518 | — | — | — | — |
| **Countries** | 37 | — | — | — | — |

Note: The random-effects specification includes a country-level random intercept and year effects. The dependent variable and continuous predictors are entered in natural logarithms. Country-clustered standard errors are reported so that inference is robust to heteroskedasticity and within-country dependence. The random-effects estimates are reported for comparison with the pooled and fixed-effects specifications.

**Table S7A.** Within- and between-country elasticity estimates for health expenditure burden and per-capita health expenditure

| Component | Population-aging elasticity of health expenditure burden | Total age dependency elasticity of health expenditure burden | Income elasticity of per-capita health expenditure |
|---|---|---|---|
| Between-country | 0.247 | 0.306 | 1.141 |
| Within-country | 0.030 | 0.197 | 0.857 |

Note: Population-aging and dependency coefficients refer to elasticities of health expenditure burden, measured as health expenditure as a percentage of GDP. The income elasticity refers to the income elasticity of per-capita health expenditure with respect to GDP per capita. For income, the estimated 95% confidence intervals are 0.974–1.307 for the between-country elasticity and 0.572–1.141 for the within-country elasticity. Because both intervals include unity, the results do not support a definitive luxury-versus-necessity classification.

Table S7B. GDP-measurement robustness using GDP per capita at PPP in constant 2021 international dollars.

| Predictor | Pooled coefficient | Pooled clustered SE | Fixed-effects coefficient | Fixed-effects clustered SE |
|---|---|---|---|---|
| Population aged 65+ | 0.280 | 0.146 | 0.192 | 0.345 |
| Total age dependency ratio | 0.378 | 0.250 | 0.215 | 0.214 |
| GDP per capita (PPP, constant 2021 international $) | 0.140 | 0.141 | −0.442 | 0.293 |

Note: This robustness specification uses World Bank indicator NY.GDP.PCAP.PP.KD (GDP per capita at PPP in constant 2021 international dollars) instead of benchmark current-US-dollar GDP while retaining the same 518 observations and log-log structure. The within-country

population-aging and total-dependency coefficients remain statistically imprecise. The within-country GDP point estimate is more negative, but the data still do not support a definitive luxury-versus-necessity classification.

Table S8A. Cluster-robust Mundlak within-between specification test.

| Test component | Result |
|---|---|
| Joint null hypothesis | Country means of ln(Pop65), ln(Dependency), and ln(GDP_pc) jointly add no explanatory information |
| Wald chi-square | 19.820 |
| Degrees of freedom | 3 |
| p-value | <0.001 |
| Interpretation | Within- and between-country relationships should be distinguished |

Note: The Mundlak/within-between formulation uses country-clustered covariance and serves as the principal specification diagnostic for within-versus-between differences under clustered inference.

Table S8B. Pesaran cross-sectional-dependence diagnostic for fixed-effects residuals.

| Component | Result |
|---|---|
| Countries | 37 |
| Annual periods | 14 |
| Country pairs | 666 |
| Pesaran CD statistic | −2.240 |
| p-value | 0.025 |

| Interpretation | Evidence of residual cross-sectional dependence |
|---|---|

Note: The significant CD statistic motivates a dependence-robust sensitivity check. The short time dimension should be considered when interpreting asymptotic panel diagnostics.

Table S8C. Driscoll-Kraay robustness for the fixed-effects specification (lag 2).

| Predictor | Coefficient | Driscoll-Kraay SE | Approx. p-value |
|---|---|---|---|
| Population aged 65+ | 0.030 | 0.043 | 0.478 |
| Total age dependency ratio | 0.197 | 0.110 | 0.074 |
| GDP per capita | −0.143 | 0.022 | <0.001 |

Note: Point estimates are unchanged from the fixed-effects model; only the covariance estimator changes. With 14 annual periods, Driscoll-Kraay inference is treated as a robustness check rather than the sole basis for inference.

Table S8D. Conventional Hausman specification test (supplementary diagnostic).

| **Test component** | **Result** |
|---|---|
| **Null hypothesis** | Random-effects estimator is consistent |
| **Alternative hypothesis** | Fixed-effects estimator is preferred |
| **Common coefficients tested** | ln(Pop65), ln(Dependency), ln(GDP_pc) |
| **Chi-square statistic** | 21.991 |
| **Degrees of freedom** | 3 |
| **p-value** | <0.001 |
| **Decision** | Reject the null hypothesis |
| **Specification implication** | Fixed effects are preferred for interpretation of within-country relationships |

Note: The conventional Hausman statistic is reported only as a supplementary diagnostic. Because substantive inference uses country-clustered standard errors, model interpretation relies principally on the cluster-robust Mundlak test in Table S8A.

**Table S9.** Heteroskedasticity diagnostic for the pooled log-log model

| **Test** | **Statistic** | **p-value** | **Decision** | **Interpretation** |
|---|---|---|---|---|
| **Breusch–Pagan LM test** | 83.543 | <0.001 | Reject homoskedasticity | Evidence of non-constant residual variance remains |

| | | | | |
|---|---|---|---|---|
| **Breusch–Pagan F test** | 6.021 | <0.001 | Reject homoskedasticity | Heteroskedasticity remains relevant after log transformation |

**Note:** The Breusch–Pagan test was applied to the pooled log-log specification including the logged core predictors and year effects. The results indicate non-constant residual variance. Accordingly, Tables S5–S7 use country-level clustered standard errors. Log transformation is used for functional-form and elasticity interpretation rather than as a remedy for all residual heteroskedasticity.

**Table S10.** Panel serial-correlation diagnostic for the log specification

| Diagnostic component | Result |
|---|---|
| **Diagnostic approach** | Wooldridge-type test for serial correlation in panel data |
| **Number of countries** | 37 |
| **Usable differenced/lagged observations** | 444 |
| **Auxiliary residual-lag coefficient** | 0.038 |
| **Clustered SE of residual-lag coefficient** | 0.047 |
| **F statistic** | 132.115 |
| **p-value** | <0.001 |
| **Decision** | Reject the null of no first-order serial correlation |
| **Inferential response** | Use country-clustered standard errors |

**Note:** The panel serial-correlation diagnostic indicates substantial within-country dependence in the error structure. Accordingly, pooled, fixed-effects, and random-effects coefficient inference uses country-clustered standard errors to account for within-country dependence. Under this Wooldridge-type auxiliary regression, the null hypothesis of no first-order serial correlation corresponds to testing the residual-lag coefficient against −0.5 rather than zero; the reported F statistic refers to this restriction.

**Table S11.** Variance inflation factor diagnostics for the logged core predictors

| Predictor | VIF | Multicollinearity interpretation |
|---|---|---|
| **ln(Population aged 65+)** | 1.523 | Low multicollinearity |
| **ln(Total age total age dependency ratio)** | 1.301 | Low multicollinearity |
| **ln(GDP per capita)** | 1.257 | Low multicollinearity |

**Note:** VIF values for the three logged continuous predictors are below commonly used diagnostic thresholds, indicating low multicollinearity among the core predictors. Derived indicators such as the Fiscal Pressure Index, public financing exposure, workforce adequacy ratio, and affordability ratio are not entered simultaneously with their parent variables in the core specification because they are deterministic transformations.

Table S12. Expanding-window validation performance including persistence and country-fixed-effects benchmarks.

| Rank | Model | Mean validation RMSE | Mean MAE | Mean sMAPE (%) | Mean $R^2$ | RMSE SD |
|---|---|---|---|---|---|---|
| **1** | One-step persistence | 0.412 | 0.327 | 3.621 | 0.957 | — |
| **2** | Country-fixed-effects linear | 0.764 | 0.614 | 7.057 | 0.888 | — |
| **3** | CatBoost Regressor | 1.238 | 0.765 | 8.155 | 0.723 | 0.254 |
| **4** | LightGBM Regressor | 1.268 | 0.788 | 8.465 | 0.706 | 0.300 |
| **5** | XGBoost Regressor | 1.425 | 0.873 | 9.465 | 0.634 | 0.240 |
| **6** | Gradient Boosting Regressor | 1.447 | 0.890 | 9.670 | 0.621 | 0.297 |
| **7** | Random Forest Regressor | 1.477 | 0.933 | 10.086 | 0.609 | 0.152 |
| **8** | Log-linear regression | 1.971 | 1.381 | 15.397 | 0.305 | 0.215 |
| **9** | Elastic Net | 2.109 | 1.552 | 17.337 | 0.205 | 0.157 |
| **10** | Linear Regression | 2.118 | 1.557 | 17.424 | 0.198 | 0.188 |

Note: Values are averages across five expanding-window validation folds (2017–2021). Each method uses only information available before the validation year. One-step persistence and the country-fixed-effects benchmark materially outperform CatBoost, showing that temporal persistence and stable country-specific expenditure levels are stronger predictors in this short annual panel than the cross-country feature-based nonlinear models.

Table S13. Fold-wise expanding-window RMSE for CatBoost and strong temporal/country-aware benchmarks.

| Fold | Training period | Training N | Validation year | Validation N | CatBoost RMSE | Persistence RMSE | Country-FE RMSE |
|---|---|---|---|---|---|---|---|
| 1 | 2010–2016 | 259 | 2017 | 37 | 1.199 | 0.197 | 0.578 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 2010–2017 | 296 | 2018 | 37 | 1.001 | 0.129 | 0.531 |
| 3 | 2010–2018 | 333 | 2019 | 37 | 1.067 | 0.219 | 0.561 |
| 4 | 2010–2019 | 370 | 2020 | 37 | 1.651 | 0.959 | 1.076 |
| 5 | 2010–2020 | 407 | 2021 | 37 | 1.270 | 0.555 | 1.075 |
| Mean | — | — | — | — | 1.238 | 0.412 | 0.764 |
| RMSE SD | — | — | — | — | 0.254 | — | — |

Note: The same chronological validation years are used for all three methods. The 2020 fold shows a pronounced increase in error for all methods, consistent with temporal instability during the COVID-19 expenditure shock. Persistence remains the strongest mean-validation benchmark across 2017–2021.

The final CatBoost configuration selected through development-period validation was: iterations = 500, learning_rate = 0.10, depth = 8, and l2_leaf_reg = 5.

Trial-by-trial provenance was not preserved sufficiently to attribute this frozen configuration uniquely to Grid Search, Random Search, Optuna TPE, or pruning/early-stopping. The documented search spaces and frozen development-period configuration are therefore reported without an unsupported attribution to a single search routine.

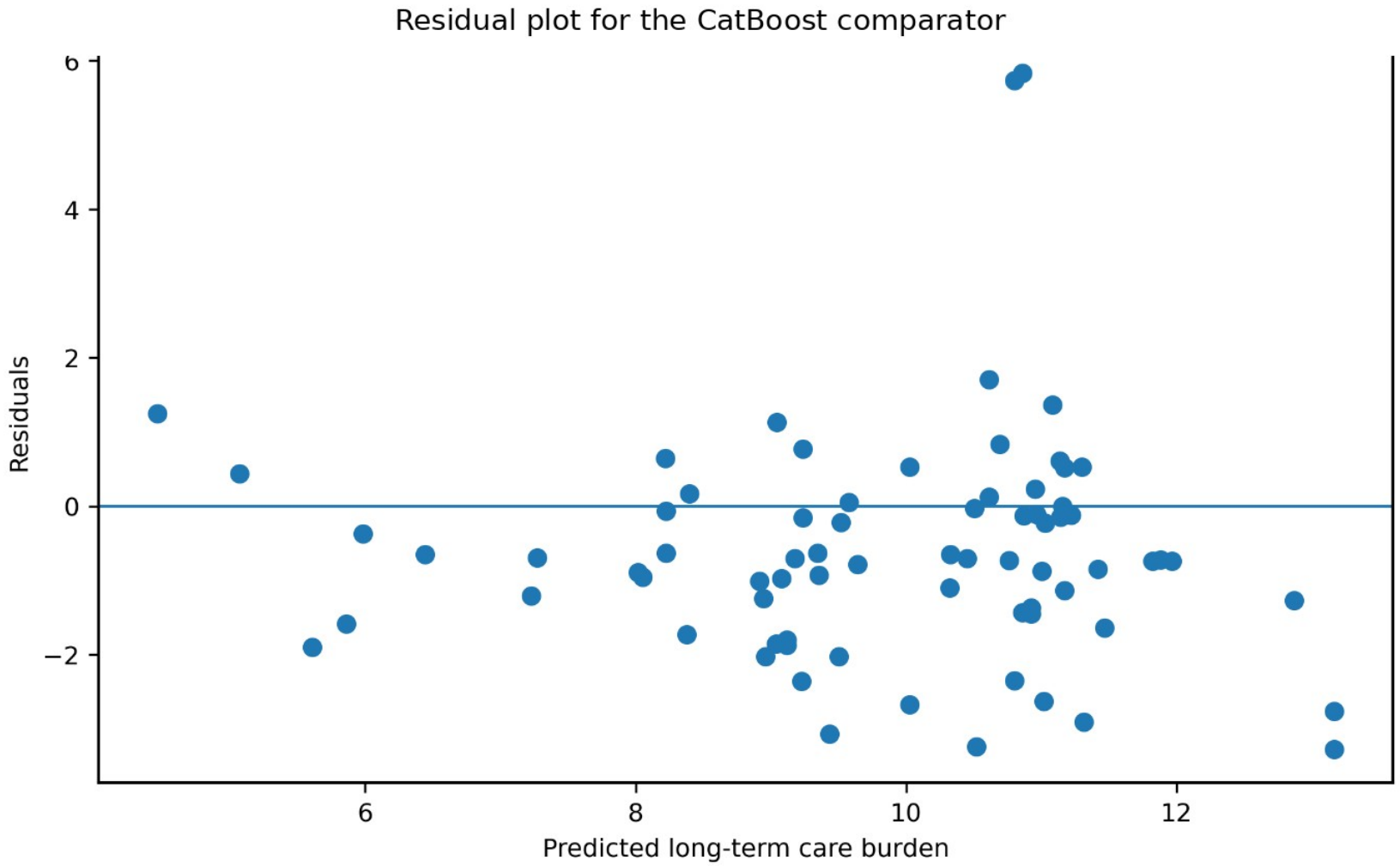



Figure S8. Residual plot for the validation-selected CatBoost comparator.

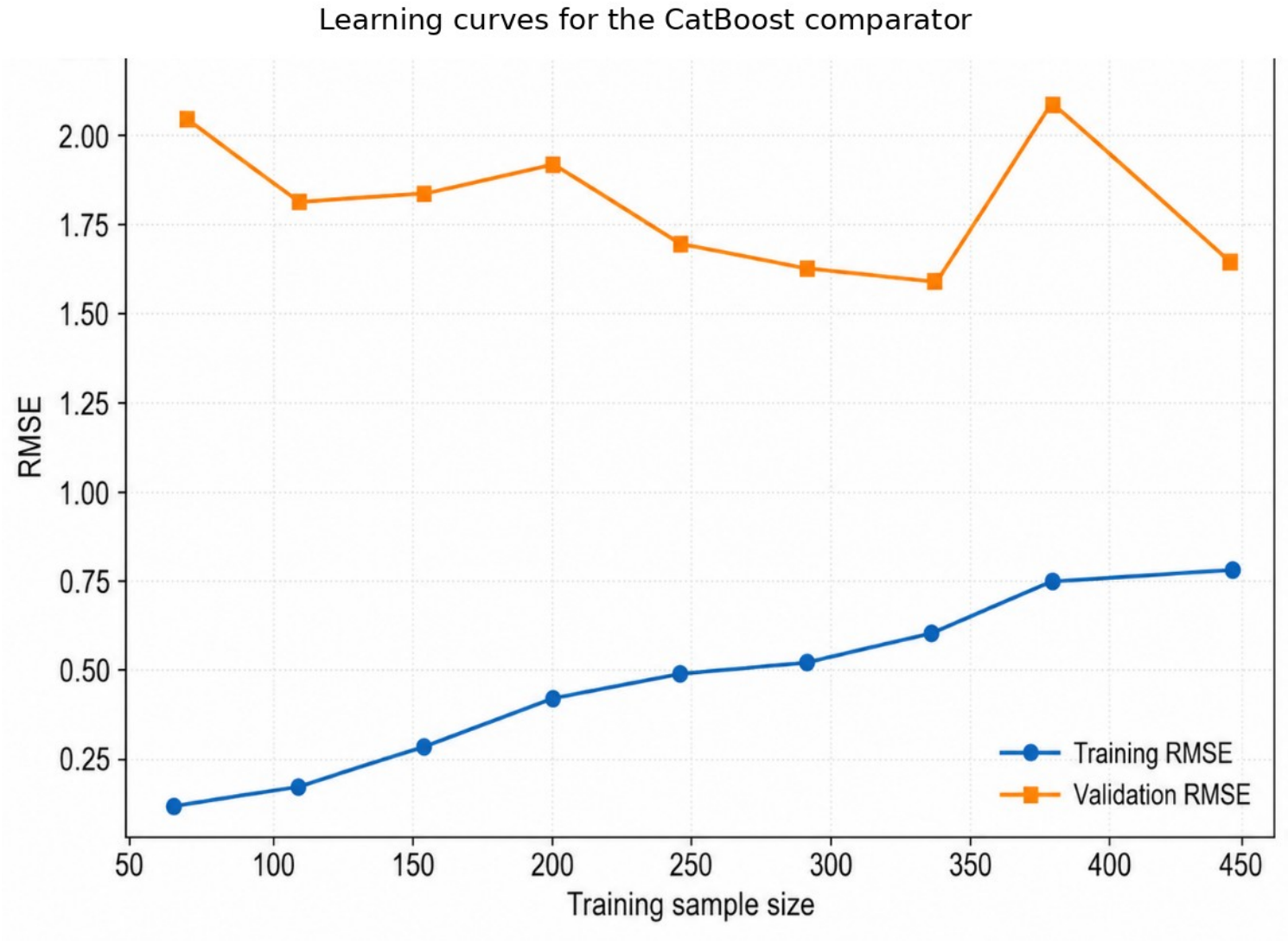



Figure S9. Learning curves for the validation-selected CatBoost comparator.

**Table S14.** Documented hyperparameter search-space design for development-period model tuning

| Model | Hyperparameter | Search values/range | Purpose |
|---|---|---|---|
| **Elastic Net** | alpha | 0.001, 0.01, 0.10, 1.0, 10.0 | Controls overall regularization strength |
| **Elastic Net** | l1_ratio | 0.10, 0.20, 0.50, 0.80, 1.00 | Controls the balance between L1 and L2 penalties |
| **Random Forest Regressor** | n_estimators | 100, 200, 300, 500 | Number of trees |
| **Random Forest Regressor** | max_depth | 3, 5, 7, 10, None | Controls maximum tree depth |
| **Random Forest Regressor** | min_samples_split | 2, 5, 10 | Minimum samples required for an internal-node split |
| **Random Forest Regressor** | min_samples_leaf | 1, 2, 4 | Minimum samples required in a terminal leaf |
| **Random Forest Regressor** | max_features | sqrt, log2, None | Number of features considered at each split |
| **XGBoost Regressor** | learning_rate | 0.01, 0.05, 0.10 | Shrinkage rate controlling boosting step size |
| **XGBoost Regressor** | max_depth | 3, 5, 7 | Maximum tree depth |
| **XGBoost Regressor** | n_estimators | 100, 200, 300 | Number of boosting rounds |
| **XGBoost Regressor** | subsample | 0.70, 0.85, 1.00 | Row-subsampling proportion |
| **XGBoost Regressor** | reg_lambda | 0, 1, 5 | L2 regularization strength |
| **XGBoost Regressor** | min_child_weight | 1, 3, 5 | Minimum child-node weight |
| **LightGBM Regressor** | learning_rate | 0.01, 0.05, 0.10 | Shrinkage rate controlling boosting step size |
| **LightGBM Regressor** | max_depth | 3, 5, 7, −1 | Tree-depth control |
| **LightGBM Regressor** | n_estimators | 100, 200, 300 | Number of boosting iterations |
| **LightGBM Regressor** | subsample | 0.70, 0.85, 1.00 | Row-subsampling proportion |
| **LightGBM Regressor** | reg_lambda | 0, 1, 5 | L2 regularization strength |
| **LightGBM Regressor** | min_child_samples | 5, 10, 20 | Minimum observations required in a leaf |
| **CatBoost Regressor** | iterations | 100, 200, 300, 500 | Number of boosting iterations |
| **CatBoost Regressor** | learning_rate | 0.01, 0.05, 0.10 | Shrinkage rate controlling boosting step size |
| **CatBoost Regressor** | depth | 4, 6, 8 | Tree depth |
| **CatBoost Regressor** | l2_leaf_reg | 1, 3, 5, 7 | L2 leaf-regularization coefficient |

Note: Hyperparameter evaluation was restricted to the 2010–2021 development period, with model assessment based on expanding-window validation. The 2022–2023 observations were reserved for independent final evaluation and were not used for model-family or hyperparameter selection. The final validation-selected CatBoost configuration was iterations = 500, learning_rate = 0.10, depth = 8, and l2_leaf_reg = 5. Gradient Boosting Regressor was included in the candidate-model comparison reported in Supplementary Table S12, but a separate Gradient Boosting hyperparameter-search space is not documented in the available tuning records and is therefore not presented in this table. RMSE = root mean squared error.

**Table S15.** Prespecified Grid Search design under expanding-window validation

| Model | Hyperparameters included | Candidate values | Total candidate combinations | Evaluation criterion | Final-test involvement |
|---|---|---|---|---|---|
| **Elastic Net** | alpha; l1_ratio | alpha: 0.001, 0.01, 0.10, 1.0, 10.0; l1_ratio: 0.10, 0.20, 0.50, 0.80, 1.00 | 25 | Lowest mean expanding-window validation RMSE | None |
| **Random Forest Regressor** | n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features | n_estimators: 100, 200, 300, 500; max_depth: 3, 5, 7, 10, None; min_samples_split: 2, 5, 10; min_samples_leaf: 1, 2, 4; max_features: sqrt, log2, None | 540 | Lowest mean expanding-window validation RMSE | None |
| **XGBoost Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight | learning_rate: 0.01, 0.05, 0.10; max_depth: 3, 5, 7; n_estimators: 100, 200, 300; subsample: 0.70, 0.85, 1.00; reg_lambda: 0, 1, 5; min_child_weight: 1, 3, 5 | 729 | Lowest mean expanding-window validation RMSE | None |
| **LightGBM Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples | learning_rate: 0.01, 0.05, 0.10; max_depth: 3, 5, 7, −1; n_estimators: 100, 200, 300; subsample: 0.70, 0.85, 1.00; reg_lambda: 0, 1, 5; min_child_samples: 5, 10, 20 | 972 | Lowest mean expanding-window validation RMSE | None |
| **CatBoost Regressor** | iterations; learning_rate; depth; l2_leaf_reg | iterations: 100, 200, 300, 500; learning_rate: 0.01, 0.05, 0.10; depth: 4, 6, 8; l2_leaf_reg: 1, 3, 5, 7 | 144 | Lowest mean expanding-window validation RMSE | None |

Note: This table documents the prespecified Grid Search parameter design used for development-period model evaluation. Expanding-window validation used 2017, 2018, 2019, 2020, and 2021 as sequential validation years, with each model trained only on observations preceding the corresponding validation year. Candidate configurations were assessed using mean validation RMSE. The 2022–2023 final test period was excluded from parameter selection and reserved for independent evaluation. Candidate-model validation performance is reported in Supplementary Table S12, fold-wise CatBoost validation results in Supplementary Table S13, and the independent final CatBoost evaluation in Supplementary Table S23.

**Table S16.** Prespecified Random Search framework under expanding-window validation

| Model | Parameters sampled | Sampling domain | Planned candidate budget | Evaluation criterion | Analytical role |
|---|---|---|---|---|---|
| **Elastic Net** | alpha; l1_ratio | Parameter domain defined in Table S14 | Up to 25 configurations | Lowest mean expanding-window validation RMSE | Regularized linear tuning benchmark |
| **Random Forest Regressor** | n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features | Parameter domain defined in Table S14 | Up to 30 randomly sampled configurations | Lowest mean expanding-window validation RMSE | Efficient exploration of nonlinear tree configurations |

| | | | | | |
|---|---|---|---|---|---|
| **XGBoost Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight | Parameter domain defined in Table S14 | Up to 30 randomly sampled configurations | Lowest mean expanding-window validation RMSE | Efficient exploration of boosting configurations |
| **LightGBM Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples | Parameter domain defined in Table S14 | Up to 30 randomly sampled configurations | Lowest mean expanding-window validation RMSE | Efficient exploration of leaf-wise boosting configurations |
| **CatBoost Regressor** | iterations; learning_rate; depth; l2_leaf_reg | Parameter domain defined in Table S14 | Up to 30 randomly sampled configurations | Lowest mean expanding-window validation RMSE | Alternative exploration of CatBoost complexity and regularization |

Note: This table documents the Random Search framework specified for development-period tuning. Candidate configurations were evaluated using the same expanding-window validation structure used for model selection, with validation years 2017–2021 and training restricted to earlier observations. The 2022–2023 final test period was excluded from all tuning decisions. Because the available tuning documentation does not identify a reproducible executed random-state value or complete candidate-by-candidate Random Search log, the table reports the prespecified sampling framework and candidate budget rather than claiming an exact trial history.

**Table S17.** Prespecified Optuna Tree-structured Parzen Estimator optimization framework

| Model | Hyperparameters considered | Search domain | Planned/allowable trial budget | Optimization objective | Validation framework | Analytical role |
|---|---|---|---|---|---|---|
| **Elastic Net** | alpha; l1_ratio | Prespecified ranges corresponding to Table S14 | 30–50 trials | Minimize mean validation RMSE | Five expanding-window validation years, 2017–2021 | Bayesian-optimization benchmark for the regularized linear model |
| **Random Forest Regressor** | n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features | Prespecified ranges corresponding to Table S14 | 50–100 trials | Minimize mean validation RMSE | Five expanding-window validation years, 2017–2021 | Bayesian exploration of nonlinear tree configurations |
| **XGBoost Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight | Prespecified ranges corresponding to Table S14 | 50–100 trials | Minimize mean validation RMSE | Five expanding-window validation years, 2017–2021 | Bayesian optimization of gradient-boosting configurations |
| **LightGBM Regressor** | learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples | Prespecified ranges corresponding to Table S14 | 50–100 trials | Minimize mean validation RMSE | Five expanding-window validation years, 2017–2021 | Bayesian optimization of leaf-wise boosting configurations |
| **CatBoost Regressor** | iterations; learning_rate; depth; l2_leaf_reg | Prespecified ranges corresponding to Table S14 | 50–100 trials | Minimize mean validation RMSE | Five expanding-window validation years, 2017–2021 | Bayesian optimization of CatBoost complexity and regularization |

Note: TPE = Tree-structured Parzen Estimator. This table documents the prespecified Optuna TPE optimization framework and allowable trial ranges considered during development-period tuning. Mean expanding-window validation RMSE was the intended optimization objective, and the 2022–2023 final test period was excluded from tuning. The reported trial ranges describe the tuning design and should not be interpreted as exact completed trial counts because complete trial-by-trial Optuna logs are not available in the current reproducibility materials. Accordingly, no convergence claim is made from this table.

**Table S18.** Prespecified pruning and early-stopping framework considered during development-period tuning

| Model | Pruning/early-stopping approach considered | Applicability | Intermediate monitoring criterion | Validation framework | Intended purpose |
|---|---|---|---|---|---|
| **XGBoost Regressor** | Validation-based pruning or model-specific early stopping during boosting | High | Validation RMSE during successive boosting iterations | Expanding-window validation within the 2010–2021 development period | Identify weak candidate configurations early and reduce unnecessary computation |
| **LightGBM Regressor** | Validation-loss-based pruning or model-specific early stopping | High | Validation RMSE during successive boosting iterations | Expanding-window validation within the 2010–2021 development period | Improve computational efficiency while limiting unnecessary boosting iterations |
| **CatBoost Regressor** | Validation-based early stopping or trial-level pruning for weak configurations | Moderate to high | Validation RMSE during boosting progress and development-period validation | Expanding-window validation within the 2010–2021 development period | Reduce unnecessary computation while retaining promising candidate configurations |
| **Random Forest Regressor** | Trial-level screening after partial candidate evaluation where computationally feasible | Limited | Interim validation RMSE | Expanding-window validation within the 2010–2021 development period | Computational screening of weak configurations |
| **Elastic Net** | No substantive pruning required | Low | Validation RMSE | Expanding-window validation within the 2010–2021 development period | Pruning offers little practical benefit for the computationally inexpensive linear model |

Note: This table documents the pruning and early-stopping framework considered for computational efficiency during development-period tuning. It does not represent a trial-by-trial execution log. Exact pruner type, stopping patience, warm-up steps, or model-specific pruning parameters should not be inferred where corresponding optimization logs are unavailable. No 2022–2023 observations were used for tuning, pruning, or early-stopping decisions. Pruning should therefore be interpreted as a development-stage computational strategy rather than an additional model-performance result.

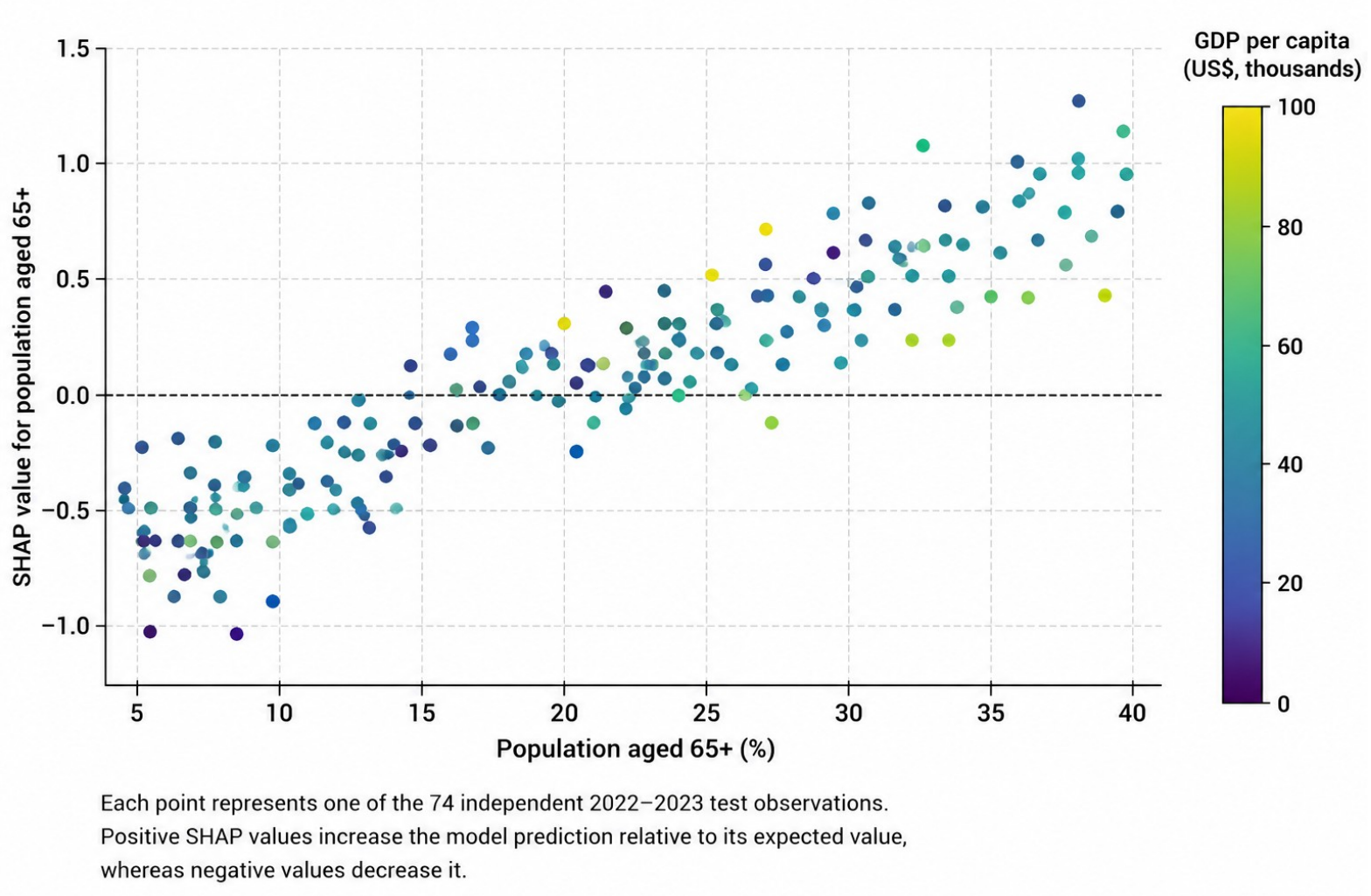



**Figure S10.** Detailed SHAP dependence of population aging in the validation-selected CatBoost model for the 2022–2023 independent test observations. Each point represents one independent test observation. The horizontal axis shows the population aged 65 years and above, the vertical axis shows its SHAP contribution to predicted health expenditure burden, and point color indicates GDP per capita. Positive SHAP values increase the model prediction relative to its expected value, whereas negative values decrease it. The pattern represents predictive model behavior and should not be interpreted as a causal effect.

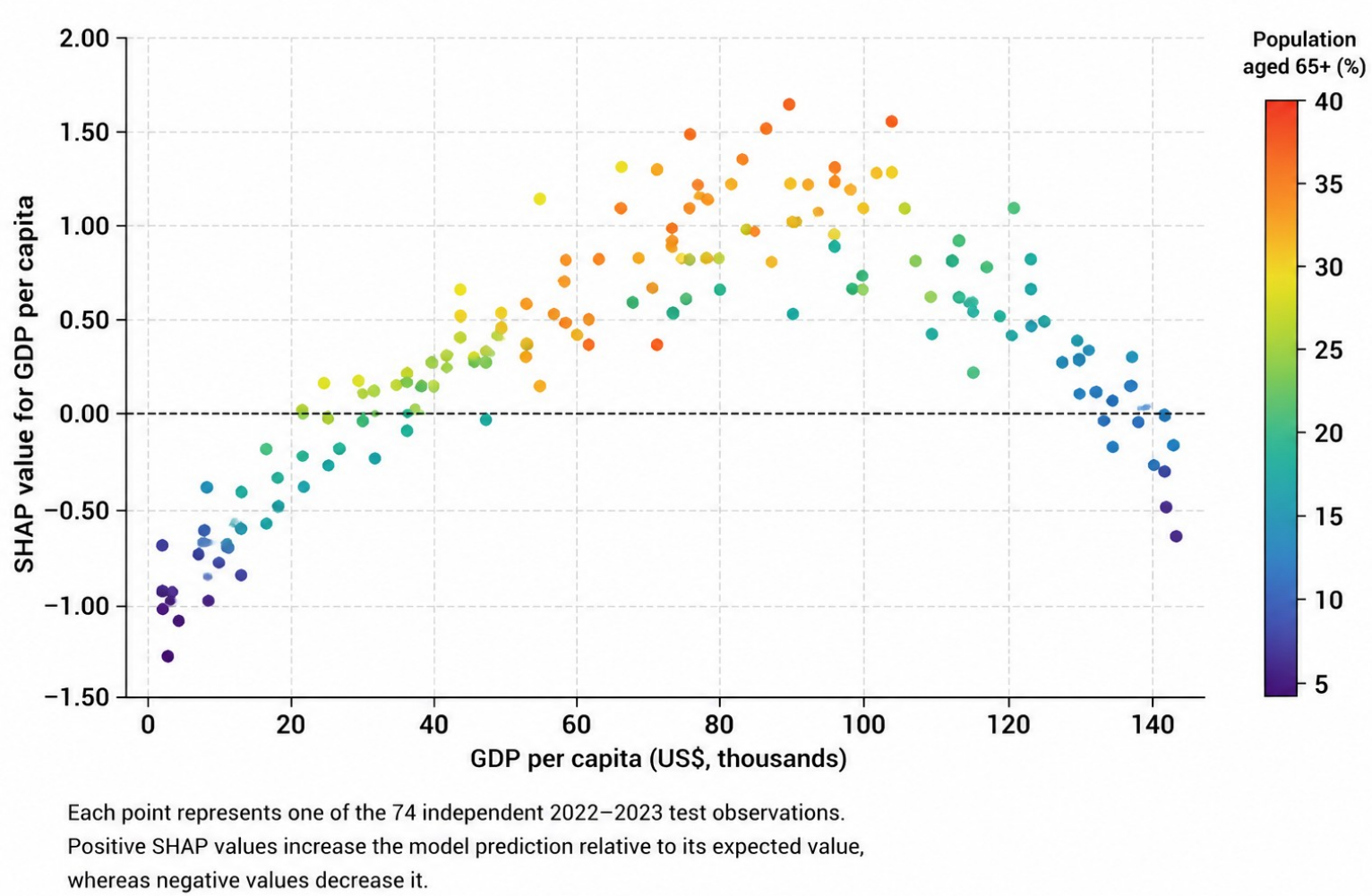



**Figure S11.** Detailed SHAP dependence of GDP per capita in the validation-selected CatBoost model for the 2022–2023 independent test observations. Each point represents one independent test observation. The horizontal axis shows GDP per capita, the vertical axis shows its SHAP contribution to predicted health expenditure burden, and point color indicates the population aged 65 years and above. Positive SHAP values increase the model prediction relative to its expected value, whereas negative values decrease it. The nonlinear pattern represents predictive model behavior and should not be interpreted as a causal income effect.

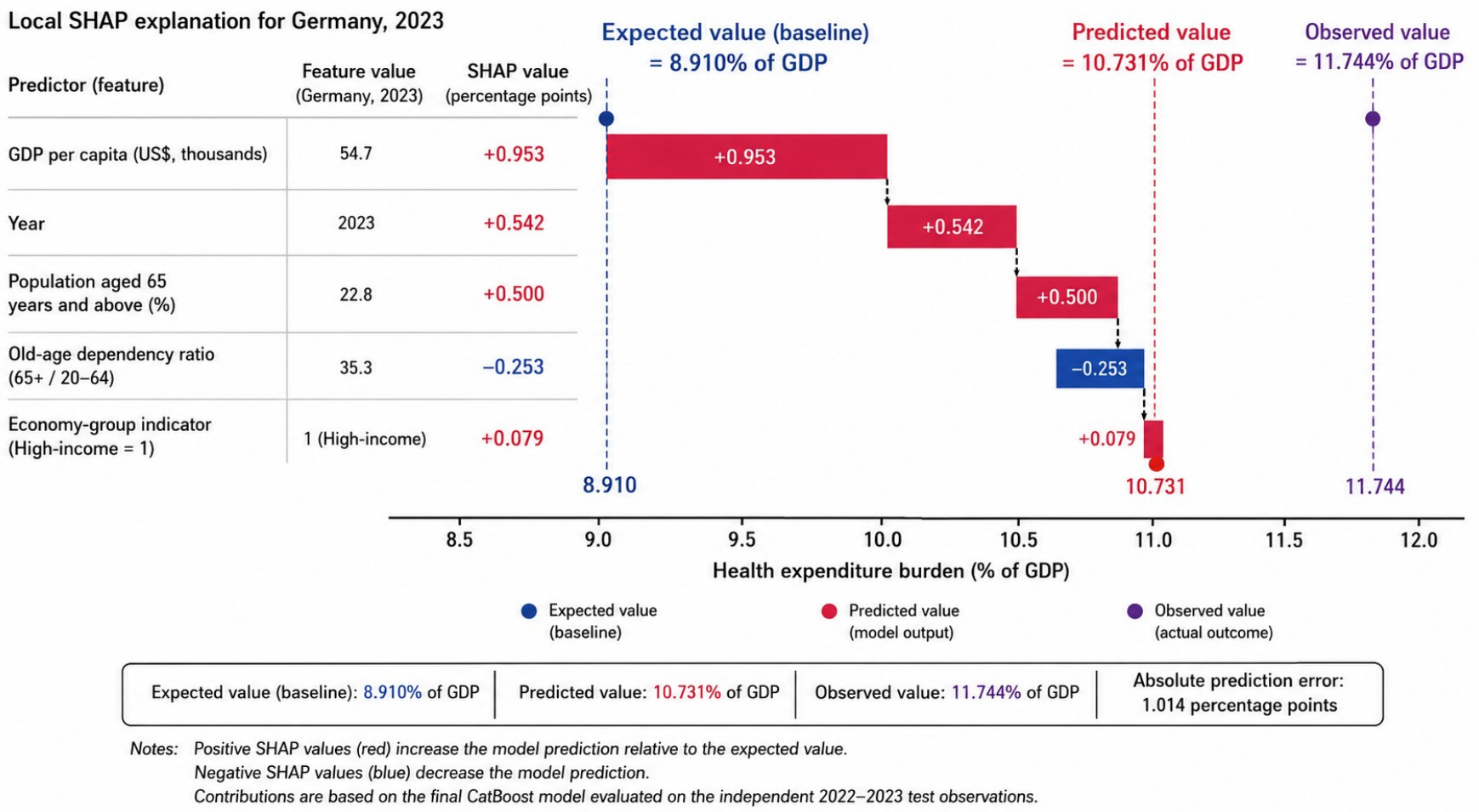



Figure S12. Local SHAP explanation for a representative independent-test observation: Germany, 2023. The observation was selected because its absolute prediction error was approximately the median among the 37 observations in the 2023 independent test year. The CatBoost model expected value was 8.910% of GDP and the predicted health expenditure burden was 10.731%, compared with an observed value of 11.744%. GDP per capita contributed +0.953 percentage points to the prediction, followed by year (+0.542), population aged 65 years and above (+0.500), total age dependency ratio (−0.253), and the economy-group indicator (+0.079). Positive SHAP values increase the model prediction relative to its expected

value, whereas negative values decrease it. The contributions describe local predictive model behavior and should not be interpreted as causal effects.

**Table S19.** Global SHAP importance of the validation-selected CatBoost model on the independent 2022–2023 test period

| Rank | Predictor | Mean absolute SHAP | SD of absolute SHAP | Share of total mean absolute SHAP (%) | Interpretation |
|---|---|---|---|---|---|
| **1** | GDP per capita | 0.843 | 0.470 | 36.1 | Largest contribution to independent-test predictions; the dependence pattern is nonlinear across the observed income distribution. |
| **2** | Population aged 65 years and above | 0.604 | 0.353 | 25.8 | Second-largest predictive contribution; higher aging values generally contribute positively to predicted health expenditure burden. |
| **3** | Year | 0.482 | 0.204 | 20.6 | Material temporal contribution to predictions across the 2022–2023 independent test period. |
| **4** | Total age dependency ratio | 0.309 | 0.267 | 13.2 | Moderate predictive contribution with a comparatively heterogeneous and non-monotonic dependence pattern. |
| **5** | Economy-group indicator | 0.098 | 0.127 | 4.2 | Small additional predictive contribution after incorporating demographic, macroeconomic, and temporal information. |

Note: SHAP values were calculated for the 74 independent 2022–2023 observations using the frozen CatBoost model. Because persistence and country-fixed-effects benchmarks outperform CatBoost, SHAP is retained only as a secondary description of CatBoost model behavior. The calendar-year feature should be interpreted cautiously because tree models partition observed year values rather than extrapolating a parametric time trend beyond training support.

Table S20. Sequential conformal performance for CatBoost and the persistence benchmark.

| Method | Evaluation subset | N | Nominal coverage | Empirical coverage | Half-width | Total width |
|---|---|---|---|---|---|---|
| CatBoost | 2022 | 37 | 90% | 86.5% | 1.821 | 3.643 |
| CatBoost | 2023 | 37 | 90% | 89.2% | 1.897 | 3.794 |
| CatBoost | Overall | 74 | 90% | 87.8% | — | — |
| Persistence | 2022 | 37 | 90% | 75.7% | 0.924 | 1.848 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Persistence | 2023 | 37 | 90% | 94.6% | 0.986 | 1.972 |
| Persistence | Overall | 74 | 90% | 85.1% | — | — |

Note: CatBoost uses expanding-window residuals from 2017–2021 as the initial conformity-score pool and updates it after observing 2022 outcomes. The persistence benchmark follows the same chronological logic using one-step persistence errors. Persistence intervals are substantially narrower, while CatBoost attains slightly higher overall empirical coverage (87.8% versus 85.1%). For persistence, 27 countries had both 2022 and 2023 outcomes covered, 9 had one of two years covered, and 1 had neither year covered. These are empirical calibration summaries under temporal and cross-country dependence, not finite-sample exchangeability guarantees.

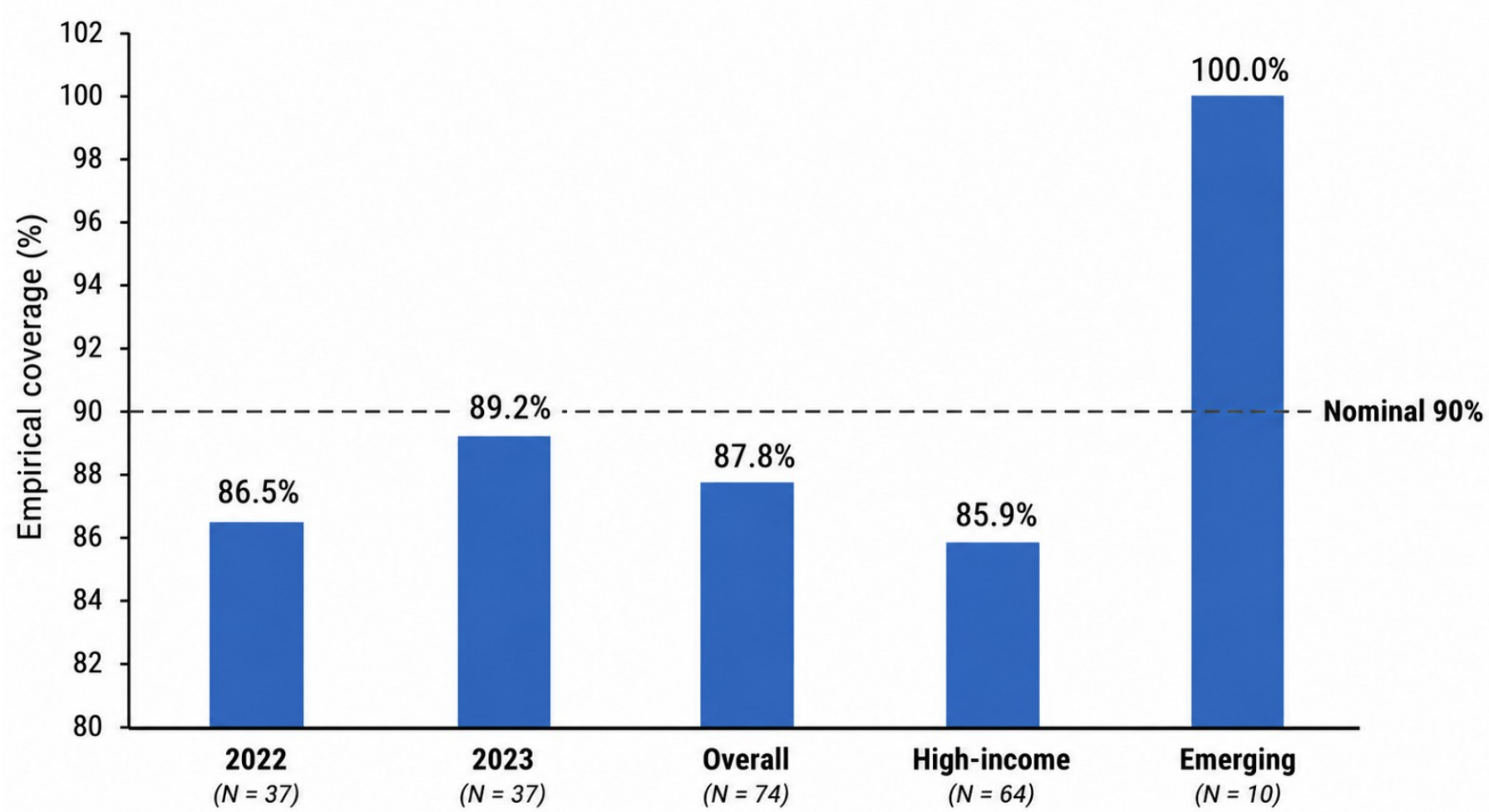



**Figure S13.** Empirical coverage of sequential 90% conformal prediction intervals. The dashed horizontal line represents the nominal 90% coverage level. Coverage is shown by evaluation year and economy group. The emerging-economy estimate is based on 10 independent-test observations and should be interpreted cautiously.

**Table S21.** Detailed assumptions for the 2024–2028 scenario sensitivity analysis.

| Scenario | Demographic trend | GDP trend | Variables manipulated | Common starting | Analytical role |
|---|---|---|---|---|---|

| | multiplier | multiplier | | point | |
|---|---|---|---|---|---|
| **Baseline** | 1.00 | 1.00 | Pop65, Dependency, GDP_pc | Observed 2023 values | Reference pathway |
| **Higher demographic change** | 1.50 for Pop65 and Dependency | 1.00 | Pop65, Dependency | Observed 2023 values | Higher demographic-pressure sensitivity; not a pure aging scenario |
| **Lower demographic change** | 0.75 for Pop65 and Dependency | 1.00 | Pop65, Dependency | Observed 2023 values | Lower demographic-pressure sensitivity; not a pure aging scenario |
| **Lower GDP growth** | 1.00 | 0.50 | GDP_pc | Observed 2023 values | Macroeconomic downside sensitivity |
| **Higher GDP growth** | 1.00 | 1.50 | GDP_pc | Observed 2023 values | Macroeconomic upside sensitivity |

Note: Country-specific historical log trends were estimated from 2010–2023 observations. The demographic sensitivities jointly scale trends in Pop65 and the total age dependency ratio. Because total age dependency includes both younger and older dependents, these are broader demographic-pressure sensitivities rather than pure aging scenarios. Government financing share and LTC workforce capacity are excluded because they are not inputs to the scenario model and have insufficient full-panel coverage.

**Table S21A.** Rolling-origin backtesting of the anchored scenario projection model.

| Forecast horizon | Number of historical origins | Scenario-model RMSE | Scenario-model MAE | Persistence RMSE | Persistence MAE |
|---|---|---|---|---|---|
| **1 year** | 7 | **0.494** | **0.393** | 0.479 | 0.381 |
| **3 years** | 5 | **0.899** | **0.740** | 0.862 | 0.712 |
| **5 years** | 3 | **0.968** | **0.761** | 0.950 | 0.765 |

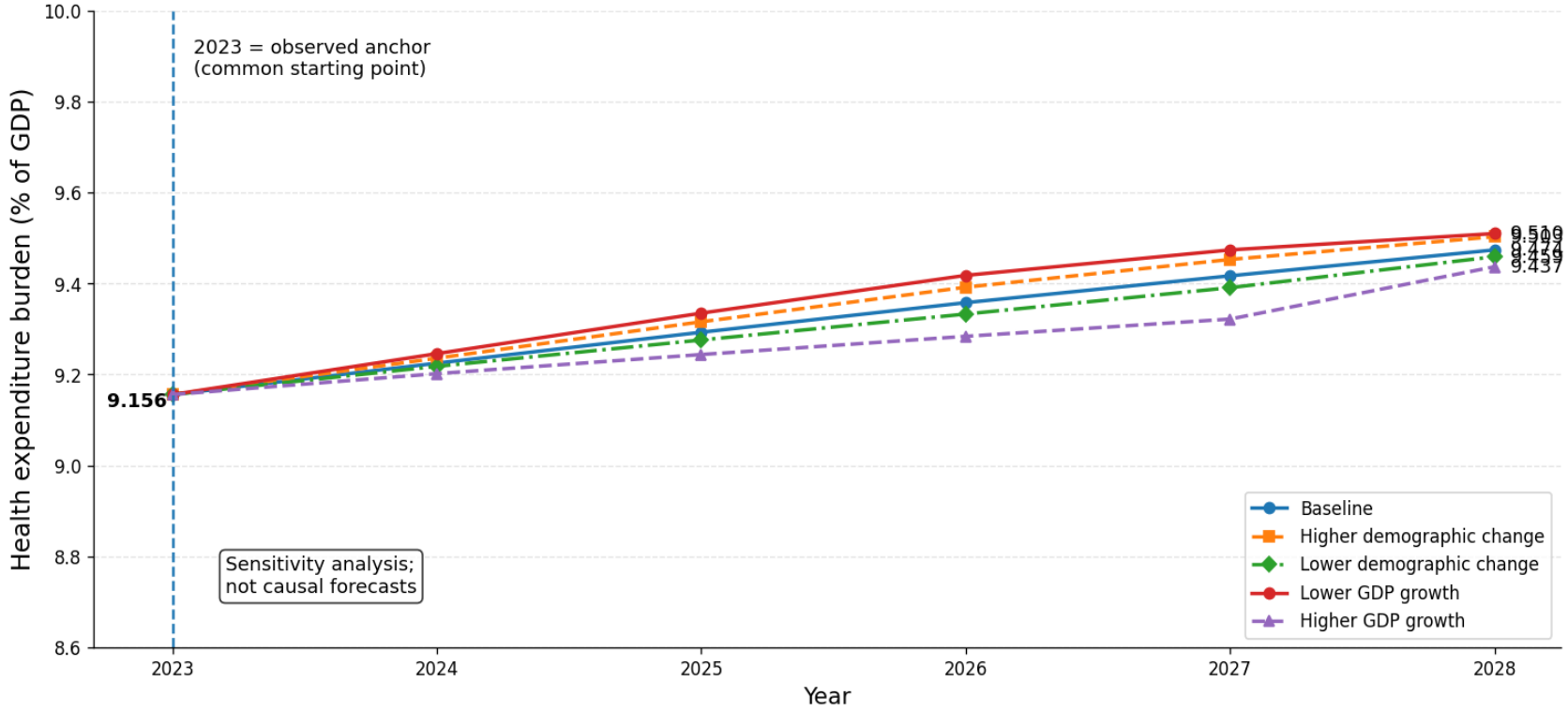
**Note:** Performance is comparable to, but does not systematically improve upon, persistence. Accordingly, the 2024–2028 outputs are interpreted as sensitivity trajectories rather than validated forecasts.

**Table S21B.** Observed predictor support and projected 2028 scenario ranges.

| Predictor | Observed 2010–2023 minimum | Observed maximum | Projected 2028 minimum across scenarios | Projected 2028 maximum across scenarios |
|---|---|---|---|---|
| **Population aged 65+** | 5.842 | 29.562 | 8.645 | 34.118 |
| **Total age total age dependency ratio** | 36.460 | 70.257 | 38.914 | 78.969 |

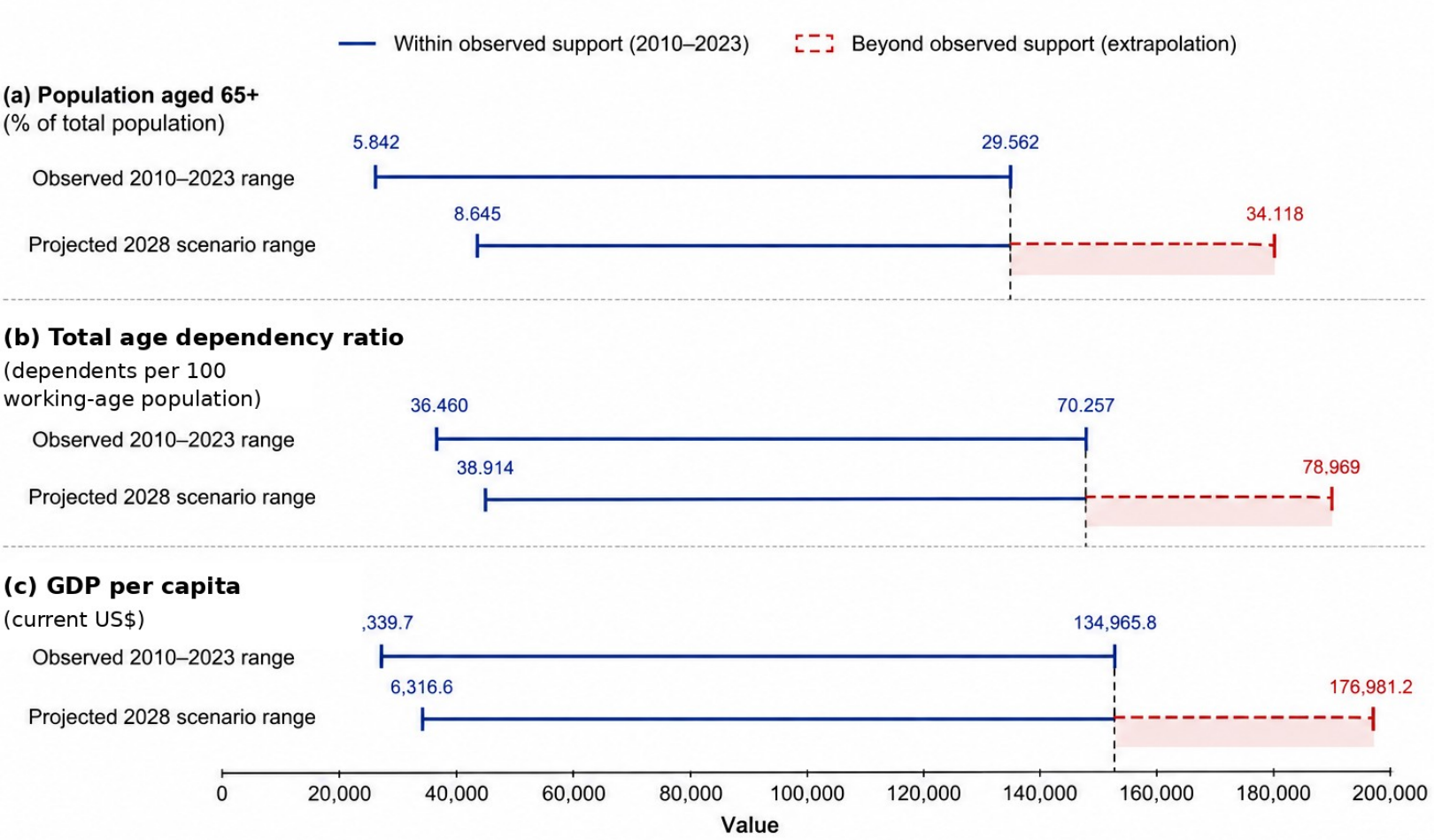
| | | | | |
|---|---|---|---|---|
| **GDP per capita** | 5,339.7 | 134,965.8 | 6,316.6 | 176,981.2 |

**Note:** Some economy-specific projected predictor values exceed the observed training support even within the five-year horizon. This is explicitly considered when interpreting the scenario trajectories and is a principal reason the analysis is described as sensitivity analysis rather than unconditional forecasting.




| Scenario | 2023 (anchor) | 2024 | 2025 | 2026 | 2027 | 2028 |
|---|---|---|---|---|---|---|
| **Baseline** | 9.156 | 9.225 | 9.293 | 9.358 | 9.417 | 9.474 |
| **Higher demographic cl** | 9.156 | 9.236 | 9.316 | 9.392 | 9.453 | 9.503 |
| **Lower demographic ch** | 9.156 | 9.218 | 9.276 | 9.333 | 9.391 | 9.459 |
| **Lower GDP growth** | 9.156 | 9.246 | 9.335 | 9.418 | 9.474 | 9.510 |
| **Higher GDP growth** | 9.156 | 9.202 | 9.244 | 9.284 | 9.322 | 9.437 |

**Figure S14.** Detailed 2023–2028 scenario sensitivity trajectories for health expenditure burden. All scenarios are anchored to the observed 2023 sample mean and diverge only according to specified demographic and GDP-growth assumptions.




**Note:** Solid blue lines indicate the observed minimum and maximum values in the 2010–2023 panel.
Projected 2028 scenario ranges are shown in blue where they fall within the observed support and in red shaded areas where they extend beyond the observed range.
Values beyond the observed support indicate extrapolation and motivate interpretation of the scenario results as sensitivity analysis rather than unconditional forecasts.

**Figure S15.** Comparison of observed predictor support and projected 2028 scenario ranges. Projected values extending beyond the observed 2010–2023 range indicate extrapolation and support interpretation of the scenario outputs as sensitivity trajectories rather than unconditional forecasts.

Table S22. Summary of robustness and sensitivity analyses, including predictive benchmarks and dependence-robust checks.

| Robustness analysis | Sample/specification | Missing-data treatment | Main analytical purpose | Interpretation |
|---|---|---|---|---|
| **R1 Core full-panel analysis** | 518 observations, 37 economies | No imputation required | Benchmark econometric and predictive analysis | Log-log estimates show weak and statistically imprecise within-country demographic effects; predictive relationships are evaluated separately |
| **R2 Core-variable specification** | HealthExp_GDP, Pop65, Dependency, GDP_pc | Complete variables retained directly | Confirms findings without sparse financing/workforce indicators | Core results are not dependent on Gov_Share or LTC_Workers |
| **R3 Restricted-sample financing description** | 41 observations, 3 countries | Missing financing observations not imputed | Exploratory financing analysis | Insufficient coverage for general cross-country financing conclusions |
| **R4 Restricted-sample workforce description** | 58 observations, 16 countries | Missing workforce observations not imputed | Exploratory workforce analysis | Provides supplementary service-capacity evidence only |
| **R5 High-income subgroup** | 32 economies | Core complete variables | Descriptive/subgroup assessment | Higher observed expenditure burden and older demographic structure; no causal interpretation |
| **R6 Emerging-economy subgroup** | 5 economies | Core complete variables | Exploratory subgroup assessment | Lower measured expenditure burden should not be interpreted as lower future care need or greater sustainability |
| **R7 Derived-indicator sensitivity** | Availability depends on parent variables | No synthetic full-panel completion | Descriptive sensitivity using FPI, affordability, public exposure, or workforce ratio | Derived indicators are deterministic transformations and are not independent evidence of fiscal sustainability |
| **R8 Scenario sensitivity** | 2024–2028 anchored trajectories | Core predictors only | Evaluate demographic and GDP-growth sensitivity | Differences among pathways are modest; backtesting does not systematically outperform persistence; outputs are sensitivity trajectories rather than validated forecasts |
| **R9** | GDP-measurement robustness | Same 518-observation panel | Uses PPP constant-2021 GDP instead of current-US$ GDP | Tests currency/price-basis sensitivity |

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|---|---|---|---|---|
| **R10** | Cluster-robust Mundlak test | Same 518-observation panel | Within-between/Mundlak formulation | Tests within-versus-between differences |
| **R11** | Cross-sectional dependence | FE residuals; 37 economies × 14 years | Pesaran CD test | Checks residual dependence across economies |
| **R12** | Dependence-robust inference | Same FE specification | Driscoll-Kraay SE, lag 2 | Sensitivity to dependence |
| **R13** | Predictive benchmarks | Identical chronological validation/test observations | One-step persistence and country-FE linear model | Tests complex ML against strong baselines |
| **R14** | Conformal benchmark | Same final-period chronology | Persistence-based sequential conformal intervals | Compares coverage and sharpness |

Note: Robustness analyses distinguish full-panel evidence from restricted-sample supplementary evidence. No missing government-financing or LTC-workforce values are imputed to create full-panel predictors. Statistical association, predictive importance, and scenario sensitivity are interpreted as distinct forms of evidence.

Table S23. Independent 2022–2023 final-test comparison of CatBoost and strong temporal/country-aware benchmarks.

| Model | Development period | Selection basis | Final test period | Test N | RMSE | MAE | sMAPE (%) | $R^2$ |
|---|---|---|---|---|---|---|---|---|
| **CatBoost** | 2010–2021 | Best of original feature-based ML candidates | 2022–2023 | 74 | 1.692 | 1.187 | 13.022 | 0.470 |
| **One-step persistence** | Chronological prior-year values | Strong temporal benchmark | 2022–2023 | 74 | 0.660 | 0.517 | 5.837 | 0.919 |
| **Country-** | Chronological | Strong country-aware linear | 2022–2023 | 74 | 0.726 | 0.580 | 6.921 | 0.902 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **fixed-effects linear** | training folds | benchmark | | | | | | |
| **Static 2021 persistence** | 2021 values only | Conservative no-2022-update benchmark | 2022–2023 | 74 | 0.801 | — | — | 0.881 |

Note: All results refer to the same 74 final-period observations. CatBoost remains a nonlinear comparator for SHAP interpretation, but it is not the preferred predictive model once persistence and country-fixed-effects benchmarks are included.

**Note:** No candidate-model ranking was performed using the final test observations. Model-family and hyperparameter selection were completed using only expanding-window validation within the development period.

**Table S24.** Reproducibility files and repository contents

| Repository component | File/content type | Purpose |
|---|---|---|
| **Cleaned analytical dataset** | CSV/Excel dataset | Provides the harmonized country-year data used for descriptive statistics, group comparisons, econometric analysis, machine-learning inputs, SHAP analysis, conformal evaluation, and scenario sensitivity. |
| **Data dictionary** | CSV/Excel table | Defines study variables, source-variable meanings, units, analytical roles, and interpretation limits. |
| **Variable-construction documentation** | Python script and supporting documentation | Documents construction of HealthExp_GDP, Public_Exposure, FPI, Worker_Ratio, and Affordability from the harmonized source variables. |
| **Missing-data diagnostics** | Tables and figures | Documents variable-level and country-year completeness, including restricted coverage of government-financing and LTC-workforce variables. |
| **Econometric analysis files** | Scripts, tables, and supporting documentation | Supports reproduction of pooled, fixed-effects, and random-effects log-log models, within-between elasticity decomposition, and associated model diagnostics. |
| **Machine-learning analysis files** | Python scripts and output tables | Supports reproduction of expanding-window validation, candidate-model comparison, final CatBoost fitting, and independent 2022–2023 test evaluation. |
| **Hyperparameter-tuning documentation** | Search-space tables and available tuning records | Documents the prespecified Grid Search, Random Search, Optuna TPE, and pruning/early-stopping frameworks used or considered within the 2010–2021 development period. Report exact trial-by-trial records only where available. |
| **SHAP explainability files** | SHAP scripts, tables, figures, and underlying test-observation data | Supports reproduction of global SHAP importance, dependence plots, interaction analysis, and local explanation for the independent 2022–2023 test observations. |
| **Conformal-prediction files** | Python scripts and output tables | Supports reproduction of the sequential 90% conformal procedure, including 2022 calibration, model refitting through 2022, 2023 calibration updating, interval widths, and empirical coverage. |
| **Scenario-sensitivity** | Scripts, tables, and figures | Supports reproduction of the anchored 2024–2028 demographic and GDP-growth sensitivity |

| | | |
|---|---|---|
| **files** | | trajectories, rolling-origin backtesting, and predictor-support diagnostics. |
| **Supplementary output files** | Word/Excel/PDF/figure files | Provides supporting tables, figures, robustness summaries, diagnostics, and methodological documentation corresponding to the Supplementary Materials. |
| **Repository documentation** | README and reproducibility notes | Describes the study scope, analytical workflow, file structure, interpretation limits, and steps required to reproduce the available analyses. |

Note: The public repository provides cleaned analytical data, a documented analytical workflow, available scripts, output tables, and supporting reproducibility materials associated with the study. The primary outcome is health expenditure burden, not direct LTC expenditure or fiscal sustainability. Government-financing and LTC-workforce variables remain restricted-coverage supplementary indicators and are not imputed to construct full-panel predictors. Hyperparameter-search tables document the prespecified development-period tuning frameworks; do not infer exact trial-by-trial tuning histories where corresponding logs are unavailable. Interpret repository outputs as authoritative only when they are generated from the documented analytical data and code and are consistent with the manuscript and Supplementary Materials.